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SIMULTANEOUS LOCALISATION OF MULTIPLE PARTIAL DISCHARGE SOURCES

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KUALA LUMPUR**

2026

**SIMULTANEOUS LOCALISATION OF MULTIPLE
PARTIAL DISCHARGE SOURCES**

CHIANG CHANG ZEE

5 **THESIS SUBMITTED IN PARTIAL FULFILMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ELECTRICAL ENGINEERING**

**DEPARTMENT OF ELECTRICAL ENGINEERING,
FACULTY OF ENGINEERING
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ABSTRACT

Partial Discharge (PD) poses a significant threat to high-voltage insulation systems, often leading to catastrophic breakdowns if undetected. While Time Difference of Arrival (TDOA) methods are effective for localizing single PD sources, real-world substation environments frequently present multiple, simultaneous PD sources that complicate detection and localization. This project proposes a comprehensive methodology to address the simultaneous localization of multiple PD sources by integrating a Digital Twin simulation environment with advanced clustering algorithms. A digital twin was developed to simulate electromagnetic (EM) waves from PD sources using Gaussian pulses and complex oscillation models, allowing for the generation of labeled datasets for algorithm testing. The study evaluates two primary feature extraction techniques: Envelope Analysis and Wavelet Decomposition (WD) combined with Principal Component Analysis (PCA). These features are processed using Particle Swarm Optimization (PSO) and Density-Based Spatial Clustering of Applications with Noise (DBSCAN) to separate and localize distinct sources. Preliminary results indicate that Envelope Analysis combined with PSO and DBSCAN achieved a 96.77% clustering accuracy on simulated waveforms, with Euclidean error of ~4m. However, experimental validation using planar monopole antennas on void and incision PD defects revealed significant challenges, particularly regarding signal attenuation and the sensitivity of WD+PCA to noise. The report concludes by identifying limitations in current feature extraction methods for real-world data and outlines future work focused on quantizing signal shape information for improved discrimination.

Keywords: Partial Discharge, Clustering, DBSCAN, PCA, Envelope Analysis.

ABSTRAK

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Nyahcas Separa (PD) menimbulkan ancaman besar kepada sistem penebat voltan tinggi, yang sering membawa kepada kerosakan katastropik jika tidak dikesan. Walaupun kaedah Perbezaan Masa Ketibaan (TDOA) berkesan untuk menyetempatan punca PD tunggal, persekitaran pencawang sebenar sering mempunyai pelbagai punca PD serentak yang merumitkan pengesanan dan penyetempatannya. Projek ini mencadangkan satu metodologi komprehensif untuk menangani penyetempatan serentak pelbagai punca PD dengan mengintegrasikan persekitaran simulasi Kembar Digital bersama algoritma pengelompokan termaju. Satu kembar digital telah dibangunkan untuk mensimulasikan gelombang elektromagnet (EM) daripada punca PD menggunakan denyutan Gaussian dan model ayunan kompleks, bagi membolehkan penghasilan set data berlabel untuk pengujian algoritma. Kajian ini menilai dua teknik pengestrakan ciri utama: Analisis Selubung dan Penguraian Wavelet (WD) yang digabungkan dengan Analisis Komponen Utama (PCA). Ciri-ciri ini diproses menggunakan Pengoptimuman Kelompok Zarah (PSO) dan Pengelompokan Ruang Berasaskan Ketumpatan Aplikasi dengan Hingar (DBSCAN) untuk memisahkan dan menyetempatan punca-punca yang berbeza. Keputusan awal menunjukkan bahawa Analisis Selubung yang digabungkan dengan PSO dan DBSCAN mencapai ketepatan pengelompokan sebanyak 96.77% pada bentuk gelombang simulasi, dengan ralat Euclidean $\sim 4m$. Walau bagaimanapun, pengesahan eksperimen menggunakan antena monopole planar pada jenis PD kecacatan rongga dan hirisan mendedahkan cabaran yang ketara, terutamanya mengenai pengecilan isyarat dan sensitiviti WD+PCA terhadap hingar. Laporan ini menyimpulkan dengan mengenal pasti batasan dalam kaedah pengestrakan ciri semasa bagi data dunia sebenar dan menggariskan kerja masa depan yang tertumpu pada pengkuantitan maklumat bentuk isyarat untuk diskriminasi yang lebih baik.

Kata kunci: Nyahcas Separa, Pengelompokan, DBSCAN, PCA, Analisis Selubung.

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ACKNOWLEDGEMENTS

First and foremost, I would like to convey my appreciation to Universiti Malaya, for providing the resources and avenues to pursue my Bachelor of Electrical Engineering.

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Next, I would also like to thank my supervisor, Ir. Dr. Wong Jee Keen Raymond, for the advice and guidance that he has provided throughout the course of this research project.

To the two PhD students who helped me set up the experiment equipment and obtain materials, I offer my heartfelt appreciation and thanks, Mr. Chan Jun Qiang and Mr. Horanapathirage Don Shiran Madhuranga. I would not have obtained meaningful real-world measurements without their generous aid.

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Next, to the two lecturers who will be moderating and evaluating this thesis, thank you for taking the time to read the work of my hands, Dr. S. M. Kayser Azam and Dr. Ahmad Haziq Aiman Bin Rosol.

To my family, friends and anyone else who has supported me in any way, whether emotional, physical or technical, thank you.

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LIST OF SYMBOLS AND ABBREVIATIONS

AC	:	Alternating Current
ADMM	:	Alternating Direction Multiplier Method
AIS	:	Air-Insulated Substation
ANN	:	Artificial Neural Network
AOA	:	Angle Of Arrival
AOK	:	Adaptive Optimal-Kernel
Avg.	:	Average
AWGN	:	Additive White Gaussian Noise
BiLSTM	:	Bidirectional Long Short-Term Memory
BPNN	:	Back Propagation Neural Network
BSS	:	Blind Source Separation
CBLAG	:	CNN-BiLSTM-Attention-GAN
CC	:	Connecting Cable
CD	:	Coupling Device
CNN	:	Convolutional Neural Network
CSA	:	Crow Search Algorithm
DBSCAN	:	Density-Based Spatial Clustering of Applications with Noise
DC	:	Direct Current
DEDO	:	Double Exponentially Damping Oscillation
DEP	:	Double Exponential Pulse
DL	:	Deep Learning
DOP	:	Damped Oscillatory Pulse
DUT	:	Device Under Test
DWT	:	Discrete Wavelet Transform

EM	:	Electromagnetic
Est.	:	Estimated
FCNN	:	Fully Connected Neural Network
FDTD	:	Finite-Difference Time Domain
FIT	:	Finite Integration Technique
F-P	:	Fabry-Perot
GAN	:	Generative Adversarial Network
GIL	:	Gas-Insulated Line
GIS	:	Gas Insulated Switchgear
GMM	:	Gaussian Mixture Model
GRU	:	Gated Recurrent Unit
GW-GAN	:	Gradient-Penalized Wasserstein GAN
HFCT	:	High-Frequency Current Transformer
HHT	:	Hilbert-Huang Transform
HV	:	High Voltage
HVDC	:	High Voltage Direct Current
KDE	:	Kernel Density Estimate
LSTM	:	Long Short-Term Memory
MI	:	Measuring Instrument
ML	:	Machine Learning
MLE	:	Maximum Likelihood Estimate
NLoS	:	Non-Line-Of-Sight
PCA	:	Principal Component Analysis
PD	:	Partial Discharge
PMA	:	Planar Monopole Antenna
PRPD	:	Phase-Resolved Partial Discharge

PSO	:	Particle Swarm Optimization
RF	:	Radio Frequency
RMSE	:	Root Mean Square Error
RSS	:	Received Signal Strength
SEDO	:	Single Exponentially Damping Oscillation
SELMS	:	Sign-Error Least Mean Square
SMG	:	Sinusoidally Modulated Gaussian
SNR	:	Signal-To-Noise Ratio
SVM	:	Support Vector Machine
SWT	:	Scatter Wavelet Transform
TDOA	:	Time Difference of Arrival
T-F	:	Time-Frequency
t-SNE	:	t-Distributed Stochastic Neighbor Embedding
UHF	:	Ultra-High Frequency
VMD	:	Variational Mode Decomposition
WD	:	Wavelet Decomposition

CHAPTER 1: INTRODUCTION

1.1 Background

Partial Discharge (PD) is a localized electrical discharge that partially bridges the insulation between conductors, due to imperfections in the insulation. Under high voltage stress, the electric field in the defect exceeds a critical threshold, causing a discharge. Over time, this damage degrades the insulation, causing void expansion and chemical breakdown, ultimately leading to complete insulation breakdown (Sharma & Bhanddakkar, 2015). The detection, and localization of PD have been under research since 1898 (Stone et al., 2022). Traditional methods of PD detection use voltage measuring devices across an impedance to detect a current spike in the device. This process is expensive and limited to devices electrically connected to the measuring instrument. Traditionally, according to IEC 60270, PD would be measured by supplying AC voltage, U , to a coupling capacitor, C_k in parallel with the test object, C_a to supply charge for the current spike. Using an impedance, Z_{mi} to convert the current spike into output voltage, it is measured via a suitable coupling device (CD), connected via a connecting cable (CC) to a measuring instrument (MI).

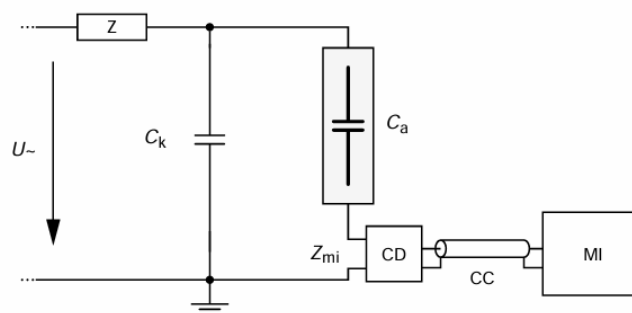


Figure 1.1: Traditional method of measuring PD occurrence (IEC60270:2025).

A high-frequency current transformer (HFCT) can also be used to obtain a voltage signal of the current spike. A newer technology is to use Ultra-High Frequency (UHF) sensors to detect electromagnetic (EM) waves emitted from the discharges. Utilizing the time difference of arrival (TDOA) of the signals at different receivers, the PD location

can be determined (Chan et al., 2025). Multiple optimization methods to localize PD from the TDOA have been developed.

The classic root finding algorithm to this problem is the Newton-Raphson method. It is a classic iterative numerical technique used to solve the non-linear system of equations derived from Time Difference of Arrival (TDOA) measurements. For a system with N sensors, the relationship between the PD source location (x, y, z) and the i -th sensor location (x_i, y_i, z_i) relative to a reference sensor (sensor 1) is given by:

$$c * \Delta t_{i,1} = R_i - R_1$$

$$R_i = \sqrt{(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2}$$

Where c is the speed of light, $\Delta t_{i,1}$ is the TDOA between sensor i and sensor 1, and R_i is the distance from the source to sensor i . To solve for (x, y, z) , the algorithm minimizes the residual error vector $\mathbf{F}(\mathbf{X})$ iteratively:

$$\mathbf{X}_{k+1} = \mathbf{X}_k - \mathbf{J}^{-1}(\mathbf{X}_k) \mathbf{F}(\mathbf{X}_k)$$

Where $\mathbf{X}_k$ is the estimated coordinate vector at iteration k , and $\mathbf{J}$ is the Jacobian matrix of partial derivatives. While precise, this method requires a good initial guess to converge and is sensitive to local minima.

To overcome the sensitivity of Newton-Raphson, Particle Swarm Optimization (PSO) is often employed. PSO is a stochastic optimization algorithm inspired by social behaviour, such as bird flocking. In this algorithm, a "swarm" of particles moves through the search space. Each particle i has a position X_i (representing potential PD coordinates) and a velocity V_i . The movement of each particle is updated based on its own best-known position (P_{best}) and the entire swarm's best-known position (G_{best}):

$$V_i^{k+1} = wV_i^k + c_1r_1(P_{best,i} - X_i^k) + c_2r_2(G_{best} - X_i^k)$$

$$X_i^{k+1} = X_i^k + V_i^{k+1}$$

Where w is the inertia weight, c_1 and c_2 are acceleration coefficients, and r_1, r_2 are random values between 0 and 1. This global search capability makes PSO robust against the non-linear errors found in real-world environments.

For detecting multiple sources, the localization process extends beyond simple equation solving into a multi-stage workflow:

- i. **Signal Acquisition:** UHF sensors capture the continuous electromagnetic spectrum.
- ii. **Pulse Isolation:** An energy-based or amplitude-based trigger isolates transient pulses from the background noise floor.
- iii. **Feature Extraction:** Each isolated pulse is transformed into a feature vector (e.g., time-equivalent bandwidth, cumulative energy, or principal components) to reduce dimensionality while retaining signal characteristics.
- iv. **Clustering:** Unsupervised algorithms (e.g., DBSCAN, K-Means) group these feature vectors. Signals in the same cluster are assumed to originate from the same PD source.
- v. **Localization:** TDOA values are calculated specifically for the centroids of these clusters, allowing for the independent localization of each unique PD source.

1.2 Problem Statement

28 In practice, there will always be multiple PD sources discharging repeatedly, overlapping, attenuating, refracting, reflecting and taking multiple paths as they travel through obstacles in a substation to reach the receiver. Signals often overlap, and higher amplitude signals can mask lower amplitude ones. In addition, PD often occurs within high voltage equipment, such as gas insulated switchgear (GIS) or transformers. Copper windings in transformers block the direct line-of-sight of UHF signals to the receiver, causing signals to take longer routes and distort the TDOA. Complex structures like GIS have different attenuation characteristics for each component within it, and are easily affected by signals from neighboring HV devices (Yizhi Fang, 2023). Therefore, a separation algorithm is needed to cluster incoming signals according to their sources for UHF radiometric detection, and to separate mixed signals successfully.

Beyond the issues of signal overlap and attenuation, the clustering of PD sources in real-world environments faces significant hurdles regarding feature robustness. In complex substations, the phenomenon of multipath propagation—where signals reflect off walls, transformers, and metallic casings—introduces "ghost" signals and distorts the pulse shape. This means that two pulses originating from the exact same PD source may look drastically different if they arrive at the sensor via different reflection paths. Consequently, standard feature extraction methods that rely heavily on pulse shape (such as Equivalent Time-Frequency bandwidth) may incorrectly classify these as different sources. Furthermore, the stochastic nature of Partial Discharge presents a challenge for unsupervised learning. PD pulses from a single defect naturally vary in amplitude and rise time due to the physics of the discharge process. If a clustering algorithm is too sensitive, it may fragment a single source into multiple clusters. Conversely, if it is too lenient, it may merge distinct sources—such as a void discharge and a nearby corona discharge—into a single cluster, rendering the subsequent localization inaccurate. This

necessitates the development of features that are invariant to propagation path distortions but sensitive to the fundamental physics of the discharge type.

Furthermore, obtaining real world PD data for development purposes poses many challenges. First, real world data would contain interference from locally transmitted radio, as well as signal attenuation (Upton et al., 2021). This would signal denoising for proper development to be done. In addition, accurately emulating PD would require a lot of resources to fabricate the PD emulator and obtain HV test equipment. Even so, emulators may not represent a wide range of possible PD types that can occur. Obtaining access to enter a substation to do testing also proves to be difficult, requiring permission from the relevant authorities. Therefore, a method to accurately simulate UHF signals obtained from PD sources in different environments needs to be developed, allowing for separation algorithms to be tested preliminarily before deploying them to real world scenarios. In fact, this field has been at the forefront of PD research for many years (Loubani et al., 2019).

1.3 Objectives

i. Data Acquisition

Develop a digital twin environment of EM wave propagation with multiple partial discharges (PD) using 3D EM wave solvers, incorporating white Gaussian noise to mimic real-world conditions. In addition, obtain real-world measured data of PD waveforms. The waveforms must be validated by showing <20% difference in clustering accuracy between real and simulated waveforms. This shows that the clustering algorithm done on simulated waveforms is a good predictor of how it will perform in a real-world scenario.

ii. Clustering of PD Sources

Evaluate and compare the performance of combinations of at least three distinct feature extraction methods (e.g., wavelet analysis, envelop detection, statistical parameters) and three separation/clustering techniques (e.g., k-means, DBSCAN, Gaussian mixture model) using the measured/simulated EM signal dataset. In addition, build and train a deep learning model for PD clustering. The goal is to achieve >80% clustering accuracy on simulated waveforms.

iii. Innovation and Optimization

Develop and implement a novel clustering method by modifying and combining existing techniques (e.g., a hybrid clustering approach). This novel method must demonstrate a quantifiable improvement over the best-performing technique from Goal 2, in terms of computational efficiency and accuracy, and at the very least achieve a target clustering accuracy of 90% or higher when performed on measured waveforms.

CHAPTER 2: LITERATURE REVIEW

2.1 PD Localization

There are several methods to localize a PD source from its emitted UHF waveforms. Time difference of arrival (TDOA) utilizes the difference in time between arriving signals at different receivers to estimate the distance of the PD from each of them. The exact moment of signal arrival is usually determined from the first peak, cumulative energy or cross-correlation plot. A set of equations that represent the relationship between TDOA and PD location is solved using optimization algorithms, such as Newton-Raphson, particle swarm optimization (PSO) and ant colony optimization (Chan et al., 2025). PSO is a classic optimization method that emulates a group of particles moving with different velocities and trying to find the global minimum of an objective function. The direction of arrival (DOA) method uses phased antenna arrays to determine which direction the PD is coming from, while the received signal strength (RSS) method uses the principle that the strength of received signal corresponds to the distance travelled by signal (Chen, 2019) (Upton et al., 2021).

In recent years, the use of advanced localization techniques for UHF based PD signals has been increasingly common. The study by Cao et al. (2023) introduces a hybrid localization method that combines time-difference-of-arrival (TDOA) and angle-of-arrival (AOA) measurements, optimized using a crow search algorithm (CSA) enhanced by particle swarm optimization (PSO). The approach addresses nonlinear equation-solving challenges in non-line-of-sight (NLoS) environments by modifying the maximum likelihood estimation fitness function and inserting initial solutions into the population to improve convergence and accuracy. Simulation experiments demonstrated that the PSO-CSA algorithm consistently outperformed other methods and showed resilience to increasing measurement errors and larger cell radii, maintaining smoother error curves and higher stability than competing methods. The PSO-CSA achieved a 3D localization

error as low as 16.64 m, compared to 130.5 m for Taylor's method, 122.9 m for Chan's algorithm, 123.9 m for hybrid TDOA/AOA, 101.8 m for PSO, 29.23 m for chaotic PSO, and 19.95 m for chaotic CSA.

Next, another study (Dhara et al., 2023) proposes a hybrid localization method that combines TDOA with relative RSS to improve accuracy in detecting PD sources within air-insulated substations (AIS). The two components are combined and interpreted via weighted least-squares optimization to obtain PD location. It was shown that the proposed method produces a 7.17% localization error, compared to pure TDOA and relative RSS methods with 11.49% and 14.4% respectively when tested under laboratory conditions. In field testing, the results were 7.92%, 11.4% and 22.36% respectively.

Another study (Lee et al., 2024) introduces a compressive spherical frequency-difference beamforming technique for localizing partial electrical discharges (PDs) in high-voltage systems, using ultrasonic emissions captured by acoustic sensor arrays. A two-step normalization process was implemented to eliminate bias from nonplanar waves, preserving only phase and relative amplitude information, while a distributed convex optimization algorithm based on the alternating direction multiplier method (ADMM) solved the localization problem efficiently. Through simulation and experimental validation, the near-field localization of PD sources was validated, with the method achieving robust performance even under limited sensor conditions.

2.2 Multiple PD Separation/Clustering

When multiple PD sources are repeatedly discharging and emitting signals, it becomes difficult to differentiate each received waveform. Multiple methods to separate and cluster signals obtained from the voltage output waveform obtained from IEC 60270/HFCT measurements have been developed. Typically, features are extracted from the signals in the time, frequency or time-frequency domain, and a clustering algorithm is then applied. The signals arriving at a sensor can either be overlapped or non-overlapped. The non-overlapped case is simpler. One example of a developed method for that case is wavelet decomposition (WD) + principal component analysis (PCA) + density-based spatial clustering of applications with noise (DBSCAN). WD passes the signal through a series of complementary high pass and low pass filters, to decompose it into detail and approximate coefficients. The signal can then be represented by scaled and shifted versions of the Mother Wavelet, a basis function defined by the user. PCA reduces the dimensionality of the sources to 3 and DBSCAN uses density to form clusters (Hao et al., 2011). Another method is to use simple geometrical structures to form envelopes of the waveform and create a time-frequency sparsity map (J. C. Chan et al., 2015). Similar methods can be found in (Wang et al., 2015), (Nimmo et al., 2017) and (Zhu et al., 2017). The overlapping case is more complex and requires blind source separation (BSS), a process to disentangle mixed signals (Tang et al., 2009). Research has shown that wavelet analysis and envelope comparison can also be used to separate sources from UHF EM waveforms (Yang & Judd, 2003) (Pinpart & Judd, 2010).

Machine learning has also been explored and applied to the field of PD clustering and separation. Applications of time-frequency analysis methods, like scatter wavelet transform (SWT) and Hilbert-Huang transform (HHT) to extract PD features, and deep learning (DL) models to cluster them are detailed by Almehdhar & Prochazka (2025). They achieved 100% experimental accuracy in PD separation. Carvalho et al. (2024)

experimented with machine learning (ML) models (K-means, gaussian mixture model, mean-shift) to cluster PD according to features such as crest factor, skewness and kurtosis, achieving 93% accuracy. (Zhang et al., 2022) and (Rauscher et al., 2024).

Among some of the more recent and advanced methods of PD clustering, Fu et al. (2024) proposes a novel approach to overcome the poor applicability of traditional PD features across sensors with different amplitude–frequency characteristics by directly extracting features from wavelet time–frequency (T–F) spectra using the t-distributed stochastic neighbor embedding (t-SNE) algorithm. By aligning waveforms and applying wavelet transforms, the dimensionality was reduced with t-SNE and the perplexity parameter was optimized using a root mean square error (RMSE)-based method. Verification across sensors with different bandwidths was done. Compared to PCA, isometric mapping, and traditional feature extraction (which used the best 2 out of 38 common PD features), the t-SNE method achieved superior separation accuracy, with clustering accuracy of 100%, although at 10 times the computational cost of the next best method.

Z. Chen et al. (2025) proposes an integrated framework for interference suppression and signal separation of multi-source partial discharge (PD) signals in HVDC gas-insulated lines (GILs), addressing the challenge of mixed PD sources and overlapping noise. Narrowband noise was suppressed in the frequency domain using a moving median threshold with Akima cubic interpolation, while white noise was reduced via a sign-error least mean square (SELMS) adaptive filter, which achieved an SNR improvement to 8.14 dB with RMSE = 0.0509. From 240 signals, eight features across time, frequency, and time–frequency domains were extracted, standardized, reduced via t-SNE, and clustered using DBSCAN, successfully separating all four discharge types without prior knowledge

of source number. Under near-realistic experimental conditions, the algorithm correctly clustered the PD signals when ≥ 160 signals were used.

2.3 Digital Twin Creation

Most attempts at creating a digital twin of PD occurrences represent partial discharge using a gaussian pulse of varying physical lengths, amplitudes and pulse widths, in software such as XFDTD, 'Ansoft' HFSS, or CST Microwave Studio. The software can numerically solve the E and H fields surrounding the PD source using the principles of finite-difference time domain (FDTD) or finite integration technique (FIT). Other attempts to model PD use equations such as single exponentially damping oscillation (SEDO), double exponentially damping oscillation (DEDO), double exponential pulse (DEP), and damped oscillatory pulse (DOP). A comparison of previous works to simulate PD is given in Table 2.1 and Table 2.2.

Table 2.1: Comparison of PD simulation techniques using software.

Reference	PD simulation technique	Purpose	What the simulation tries to represent	Findings (related to simulation)
(He et al., 2020)	Gaussian current impulse in Ansoft HFSS, with added white noise.	Separating multisource PD to localize.	UHF PD signals	Good agreement between simulation and measurement results
(Loubani et al., 2019)	Gaussian current impulse in XFDTD	Analyzing propagation characteristics of PD.		-
(Qin et al., 2020)		Mixing characteristics of PD signals in GIS		Verified PD mixing characteristics
(Tian & Kawada, 2014)		Estimating DOA of PD		-
(Zheng et al., 2023)	Dielectric Barrier Discharge model + FDTD	EM wave transmission in GIS.		-
(Umemoto & Tenbohlen, 2018)	CST to model EM wave propagation of antenna emitted signal.	Comparing simulated and measured PD signal propagation in transformer		Similarly shaped simulated and measured signals.
(Toshihiro et al., 2008)	Gaussian current impulse in FDTD	Comparing simulated and measured propagation of EM waves in GIS		-
(Ohtsuka et al., 2006)	Geometrically shaped current impulse in FDTD	Estimating PD charge quantity from signals detected in GIS		-
(Jaber et al., 2016)	Gaussian current impulse in CST	Compare PD emulator pulse with simulated.		Good agreement between simulated and measured signals
(Zanjani et al., 2012)		Analyze EM wave propagation in distribution transformer.		

Table 2.2: Comparison of PD simulation techniques using equations.

Reference	PD simulation technique	Purpose	What the simulation tries to represent	Findings (related to simulation)
(Lin et al., 2022)	DEDO & SEDO + Gaussian white noise	Denoising	Acoustic PD emissions	-
(Ma et al., 2002)	DEP, DOP	Exploration of wavelet analysis on PD detection	Current in the detection circuit caused by PD	-
(Sahnoune et al., 2024)	DEP, DOP	Denoising	Current in the detection circuit caused by PD	-
(Kumar et al., 2023)	Mixture of SEDO + DEP	Denoising	Acoustic PD emissions	-
(Friebe & Jenau, 2024)	Mixture of SEDO + DEP	Denoising	UHF PD signals	Simulated waveform made to closely resemble measured signals

Zheng et al. (2023) and Ohtsuka et al. (2006) were the only ones that didn't use Gaussian current pulses to model PD. Most of the simulations were done with the purpose of validating the accuracy of simulated single PD waveforms, compared to experimental measured waveforms. Only (Yun et al., 2020) and (He et al., 2020) investigated the simulation of multiple PD sources for the purpose of applying separation/clustering algorithms to them. Research regarding PD simulation using equations like SEDO, DEDO, DEP and DOP are not specifically done for PD clustering either. In fact, some of them do not even focus on modelling UHF emissions, but rather acoustic waves or current pulses. A clear research gap exists in this area, because more research is needed to confirm the eligibility of testing separation algorithms on simulated PD waveforms. Verification needs to be done on whether a simulation would be capable of producing different types of waveforms that accurately resemble experimentally collected samples, with extractable features that are unique to each different simulated PD.

2.4 Deep Learning (DL) Applications in PD Clustering

Beyond traditional statistical analysis, recent advancements have increasingly leveraged Deep Learning (DL) for the automated feature extraction of PD signals. While manual feature engineering relies on calculating specific metrics like skewness and kurtosis, DL models, particularly Convolutional Neural Networks (CNNs), are designed to process grid-like topology data such as time-series waveforms or Phase-Resolved Partial Discharge (PRPD) patterns. By utilizing learnable filters that scan input data, CNNs can detect local patterns like sharp rise times or oscillatory decays that are characteristic of specific PD types. For instance, Chang et al. (2022) demonstrated that applying CNNs to pulse sequence data significantly improved pattern recognition accuracy in underground cable joints by effectively capturing spatial-temporal features that traditional methods often miss. Similarly, Jiang et al. (2023) utilized a stepwise transfer learning approach with CNNs to enhance PD pattern recognition in generator stators, achieving high accuracy even with limited training samples. This shift towards automated feature extraction represents a significant evolution in PD diagnostics, reducing the reliance on expert domain knowledge for signal processing.

Another study (Zhang et al., 2022) presents a partial discharge (PD) pattern recognition method for oil-immersed transformers using a multifrequency fiber-optic Fabry–Perot (F–P) ultrasonic sensing array, adaptive optimal-kernel (AOK) time–frequency representation, and a deep learning framework (ResNet-18). Three F–P probes with resonant frequencies at 41, 75, and 112 kHz were deployed to capture PD signals across different bands, improving sensitivity and interference immunity compared to conventional piezoelectric sensors. The AOK method was used to generate high-resolution time–frequency matrices (300×375) that highlighted attenuation effects, especially for high-frequency components blocked by solid insulation. A database of 1,100 samples (220 per PD type) was used to train the modified ResNet-18, achieving

98% recognition accuracy across five PD types, outperforming piezoelectric sensor-based detection (92.9%). Confidence analysis showed probabilities >0.99 for most known PD types, though signals attenuated by pressboards and windings had lower diagnostic confidence, while unknown PD or noise signals produced dispersed or low confidence predictions.

The study by Rauscher et al. (2024) introduces a systematic framework for improving PD versus noise classification in end-of-line stator testing under surge voltage. Six networks: fully connected neural networks (FCNNs), 1D/2D convoluted neural networks (CNNs), long-short term memory (LSTM), bidirectional LSTM (BiLSTM), and gated recurrent unit (GRU), were combined with different time-frequency transforms and were optimized via hyperparameter tuning. Results showed that a 2D CNN with continuous wavelet transform achieved the highest accuracy of 99.75% on unseen test objects, with detection rates above 95% even for PD signals near the noise floor. Receiver operating characteristic analysis confirmed a true positive rate $>99.5\%$ and false positive rate $<0.14\%$, outperforming conventional threshold-based detection. The concepts of PD versus noise separation detailed here can be applied to PD-PD separation.

L. Chen et al. (2025) proposes a method to classify PD acoustic emissions. First, Adaptive Variational Mode Decomposition (VMD) that uses sample entropy optimization, processes the signal by automatically selecting the number of modes and penalty factors. Next, a hybrid deep learning model (CNN + LSTM) is trained. It consists of a 1D-CNN branch that extracts spatial features (pulse shapes, high-frequency discharge signatures), and a Bidirectional LSTM branch that captures sequential dependencies in discharge activities. Together, they learn both transient discharge morphologies and their evolutionary patterns. The method achieved 96.2% classification accuracy for multi-source defects ($\approx 38\%$ better than wavelet methods), 5.8 mm mean localization error ($\approx$

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53% better than traditional TDOA) and 4.2° angular accuracy even under high noise. The processing time was ~0.8s, making it practical for near real-time monitoring.

Another study by Ding et al. (2025) compresses CNN models for transformer partial discharge (PD) diagnosis using reinforcement learning-guided filter pruning via geometric median, achieving 85–96% parameter reduction with accuracy losses under 1%. Deployment tests on Raspberry Pi 4B and Jetson Nano confirmed real-time feasibility, with power consumption reduced by ~40% and memory usage halved. While the paper does not address PD source separation directly, its results show that compressed CNNs can still classify UHF PD signals reliably.

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Fu et al. (2025) propose a transformer partial discharge (PD) recognition framework that integrates convolutional neural networks (CNN), bidirectional long short-term memory (BiLSTM), and an attention mechanism, enhanced with a gradient-penalized Wasserstein GAN (GW-GAN) for data augmentation. Using wavelet time–frequency spectrograms of PD signals, CNN extracts local features, BiLSTM captures bidirectional temporal dependencies, and the attention mechanism adaptively weights features to improve classification. To address data imbalance, GW-GAN generated realistic synthetic samples, boosting generalization. Results showed that the CNN-BiLSTM-attention-GAN (CBLAG) model achieved a recognition accuracy of 97.36%, outperforming CNN+CGAN by 1.8% and traditional SVM by 5.8%. The CBLAG model reached the highest overall accuracy (98.8%) among six tested methods, while baseline models like BPNN lagged at 64.8%.

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2.5 Research Gap

Despite the extensive literature on PD detection, significant gaps remain in the development of high-quality Digital Twin environments for multiple-source scenarios. While previous studies have successfully simulated single PD sources using Gaussian pulses, there is a lack of comprehensive research that models the simultaneous propagation of multiple distinct PD types (e.g., void, surface, etc) within a 3D space, for the specific purpose of clustering. For clustering, the PD pulse shape is important, as simulating the wrong pulse shape would lead to discrepancies from measured waveforms. Current research that simulates PD for other purposes, i.e. TDOA estimation, propagation characteristics, denoising, do not place importance on how accurate the shape of the pulse is, because the shape is redundant for those applications.

Furthermore, existing clustering methodologies face a trade-off between computational efficiency and accuracy. Advanced manifold learning techniques like t-SNE offer high separation accuracy but are computationally expensive and unsuitable for real-time monitoring systems. Conversely, faster methods like K-means often fail when clusters are of arbitrary shape or varying densities, which is common in PD data. There is a distinct need for an optimized clustering framework that integrates robust feature extraction—capable of handling signal attenuation and noise—with an efficient density-based algorithm (like DBSCAN) that operates independently of the noise floor. This research aims to bridge this gap by validating low-latency clustering algorithms against both a high-quality Digital Twin and experimental measurements.

CHAPTER 3: METHODOLOGY

3.1 Digital Twin Simulation Method

3.1.1 COMSOL Simulation

A digital twin environment would be created to generate data for clustering analysis. Two different methods of modelling a PD occurrence will be explored. The first is to use a gaussian current pulse and simulate the EM waves propagating from it:

$$I(t) = I_{max}e^{-t^2/2\sigma^2} \quad (1)$$

Where I_{max} is maximum current amplitude, σ is the pulse width (Loubani et al., 2019).

The Gaussian pulse can be modelled in COMSOL Multiphysics using the RF Module, specifically utilizing the Electromagnetic Waves, Transient (temw) interface. This interface solves the time-dependent Maxwell's equations, which is essential for visualizing the propagation of electromagnetic pulses in the time domain. The simulation environment can be established by defining a 3D air box with relative permittivity $\epsilon_r \approx 1$ to represent the substation space. The PD source can then be defined using a Lumped Port boundary condition applied to a dipole geometry. The input for this source is governed by the Gaussian pulse Equation (1), entered as an analytic function within the software. A physics-controlled mesh can be generated, ensuring the maximum element size is smaller than the shortest wavelength of the pulse to prevent numerical dispersion. The time-dependent solver is then configured with picosecond-range time steps to resolve the Ultra-High Frequency (UHF) components of the discharge. Finally, electric field probes are placed at specific coordinates matching the experimental sensor layout to record the voltage variations over time.

3.1.2 Waveform Mixing

The second method is to simulate a PD emitted EM wave as a combination of multiple basic waveforms. Different types of PD contain superpositions of different base

waveforms, of various time delays and amplitudes. The waves are assumed to emit from a specified point in space and attenuate based on how far they travel to each receiver.

Gaussian noise is added to the received signal to model real-world conditions.

Several base waveforms used in the modelling are given below:

1. Single Exponentially Damping Oscillation (SEDO)

$$V_0 e^{-\frac{(t-t_d)}{\tau}} \sin(2\pi f_c(t - t_d) + \theta) \quad (2)$$

Where V_0 is peak amplitude, t_d is time delay, τ is decay time constant, f_c is center frequency of oscillation, θ phase offset. The values of the signal for $t - t_d < 0$ are forced to zero, and this is done for all subsequent signals. This is a modified version of the equation used in (Lin et al., 2022).

2. Double Exponentially Damping Oscillation (DEDO)

$$V_0 (e^{-\alpha(t-t_d)} - e^{-\beta(t-t_d)}) \sin(2\pi f_c(t - t_d) + \theta) \quad (3)$$

Where α and β control the rise and fall times, where $\alpha < \beta$ (Lin et al., 2022). This model better represents what a PD EM wave would look like, containing a voltage rise and fall, with oscillations.

3. Double Exponential Pulse (DEP)

$$V_0 (e^{-\alpha(t-t_d)} - e^{-\beta(t-t_d)}) \quad (4)$$

This is simply a DEDO without the oscillations, as introduced and used in (Ma et al., 2002).

4. Sinusoidally Modulated Gaussian (SMG)

$$Ae^{-\frac{((t-t_0-t_d)^2)}{2(\tau^2)}} \sin(2\pi f_c(t-t_0-t_d)) \quad (5)$$

A is the amplitude, t_0 is the pulse center. This signal has an equal rise and fall time, and is symmetrical about t_0 (Miljanovic et al., 2021).

Six different PD sources were simulated with Python, with the parameters as shown in Table 3.1. The sampling rate is set to be 5GSs^{-1} . The values are in the format of $N(\mu, \sigma)$, where N is a normal distribution with mean μ and standard deviation σ .

Table 3.1: Parameters used for PD Simulation.

PD1	DEDO	$V_0 \sim N(2, 0.8)$ $\alpha \sim N(1e7, 0.4e7)$ $\beta \sim N(1e8, 0.4e8)$ $f_c \sim N(1e7, 0.4e7)$ $\theta = 0$ $t_d = 0$	+	SMG	$A \sim N(0.7, 0.28)$ $t_0 = 1.5e-7$ $\tau \sim N(0.4e-7, 0.16e-7)$ $f_c \sim N(0.5e7, 0.2e7)$ $t_d = 2e-7$			
PD2	SEDO	$V_0 \sim N(3, 1.2)$ $\tau \sim N(0.1e-6, 0.4e-7)$ $f_c \sim N(2.3e7, 0.92e7)$ $\theta = 0$ $t_d = 0$	+	DEDO	$V_0 \sim N(0.9, 0.36)$ $\alpha \sim N(1e7, 0.4e7)$ $\beta \sim N(1e9, 0.4e9)$ $f_c \sim N(1.7e7, 0.68e7)$ $\theta = 0$ $t_d = 1.8e-7$			
PD3	SEDO	$V_0 \sim N(0.5, 0.2)$ $\tau = \infty$ $f_c = 0$ $\theta = \pi/2$ $t_d = 0$	-	SEDO	$V_0 \sim N(0.5, 0.2)$ $\tau \sim N(0.1e-6, 0.4e-7)$ $f_c = 0$ $\theta = \pi/2$ $t_d = 0$	×	SEDO	$V_0 \sim N(9, 3.6)$ $\tau \sim N(0.15e-6, 0.6e-7)$ $f_c = 1.68e7$ $\theta = 0$ $t_d = 0$
PD4	DEP	$V_0 \sim N(3, 1.2)$ $\alpha \sim N(5e6, 2e5)$ $\beta \sim N(1.5e7, 6e6)$ $t_d = 1.88e-7$	×	SEDO	$V_0 \sim N(1.5, 0.6)$ $\tau \sim N(0.18e-6, 0.72e-7)$ $f_c \sim N(0.9e7, 0.36e6)$ $\theta = 0$ $t_d = 1.88e-7$	+	SEDO	$V_0 \sim N(3.8, 1.52)$ $\tau \sim N(0.12e-6, 0.48e-7)$ $f_c \sim N(1.7e7, 0.68e7)$ $\theta = 0$ $t_d = 0$
PD5	SMG	$A \sim N(1.2, 0.48)$ $t_0 = 1e-7$ $\tau \sim N(0.2e-7, 0.08e-7)$ $f_c \sim N(2.5e7, 1e6)$ $t_d = 0$	+	SMG	$A \sim N(0.2, 0.08)$ $t_0 = 2.5e-7$ $\tau \sim N(0.2e-7, 0.08e-7)$ $f_c \sim N(2.5e7, 1e6)$ $t_d = 0$			
PD6	DEDO	$V_0 \sim N(2.5, 1)$ $\alpha \sim N(1.5e7, 0.6e7)$ $\beta \sim N(1.5e8, 0.6e8)$ $f_c \sim N(1.5e7, 0.6e7)$ $\theta = 0$ $t_d = 0.5e-7$	+	SMG	$A \sim N(0.3, 0.12)$ $t_0 = 3.5e-7$ $\tau \sim N(0.3e-7, 0.12e-7)$ $f_c \sim N(3e7, 1.2e6)$ $t_d = 0$			

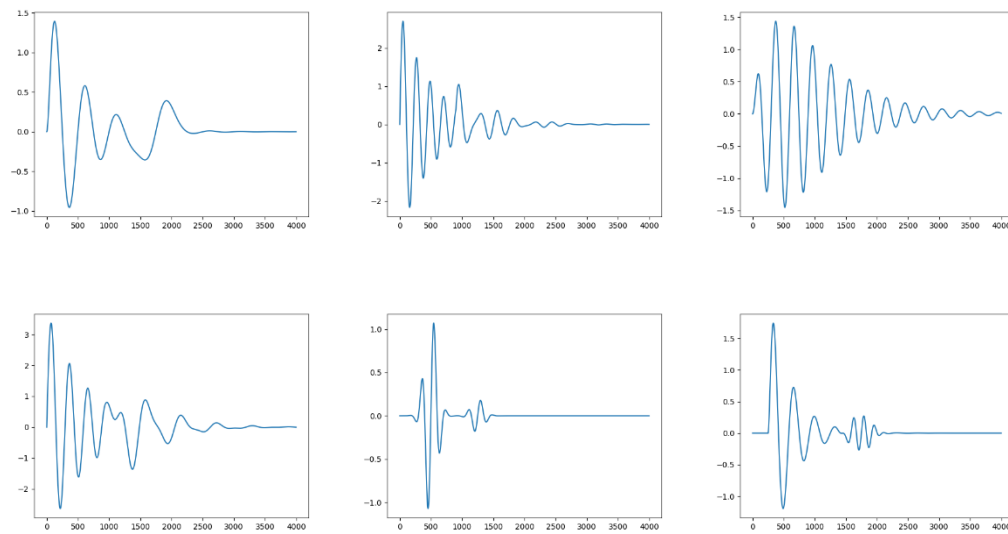


Figure 3.1: The generated waveforms, in order from 1 to 6.

Each PD source is given a different geometrical coordinate, as shown in Table 3.2.

Table 3.2: PD source locations.

PD Source	Coordinate (m)
PD1	1.2, 2.3, 3.1
PD2	2.9, 6.3, 5.4
PD3	4.5, 7.8, 6.2
PD4	6.1, 1.7, 3.5
PD5	7.8, 1.2, 4.9
PD6	9.2, 3.5, 2.8

The simulated sensor coordinates are [1,9,1.8], [1,7,1.4], [1,5,1.8], [1,3,1.4] for sensor 4 to 1 in that order. They are arranged at different x, y positions, and different heights, z, which improve precision and accuracy as compared to layouts with all the sensors at the same height (Chan et al., 2026). Each PD is set to discharge 8 times, uniformly distributed in a time span of 0.4ms. Each PD waveform would then be added to the simulated sensor output signal, with the appropriate time difference of arrival (TDOA) for each sensor. The amplitude of the signal detected at each sensor was scaled by the factor $e^{-0.2l}$, where l is the distance between PD source and the sensor to account for attenuation.

Additive white gaussian noise (AWGN) was added at $SNR = 20 \text{ dB}$ per channel, with the noise power in watts as the variance of the noise signal being added. The final simulated sensor outputs consist of a series of PD waveforms at different times, from different sources and with different amplitudes. An example of the output of sensor 1 (Channel 1) is shown in Figure 3.2. The term ‘sensor’ is synonymous with ‘channel’ in this report, because the sensor output is detected by the channels of an oscilloscope.

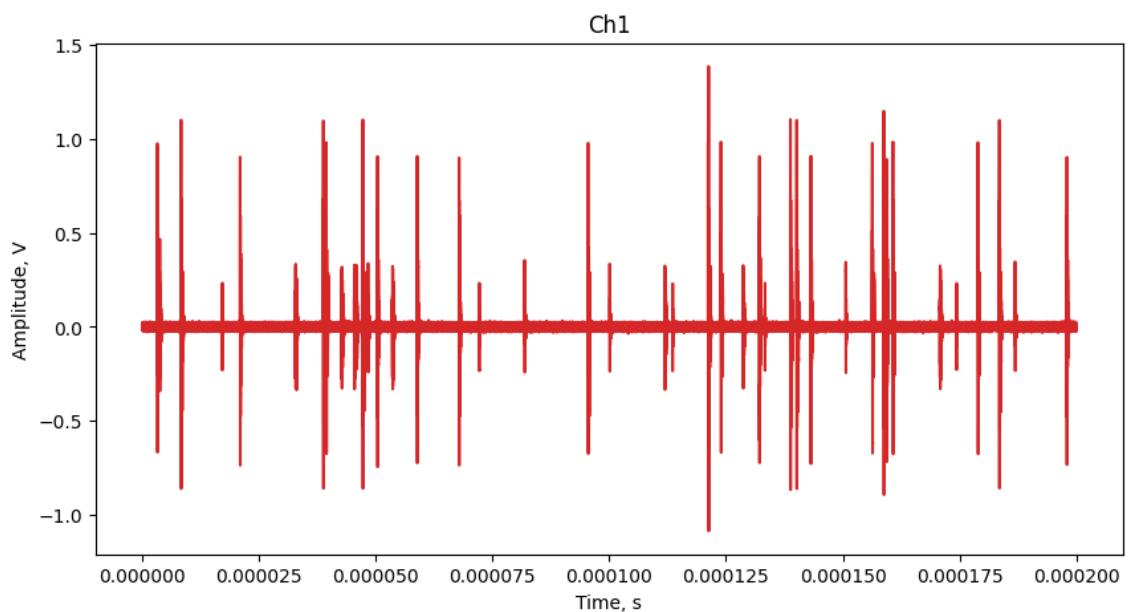


Figure 3.2: The simulated output of sensor 1 (Ch1).

3.2 PD Waveform Isolation

For clustering to be performed on actual measurements of PD occurrences, an algorithm needs to be developed to isolate portions of the real-world sensor outputs that contain PD for further processing. The workflow of isolating portions of the signal is shown in Figure 3.3.

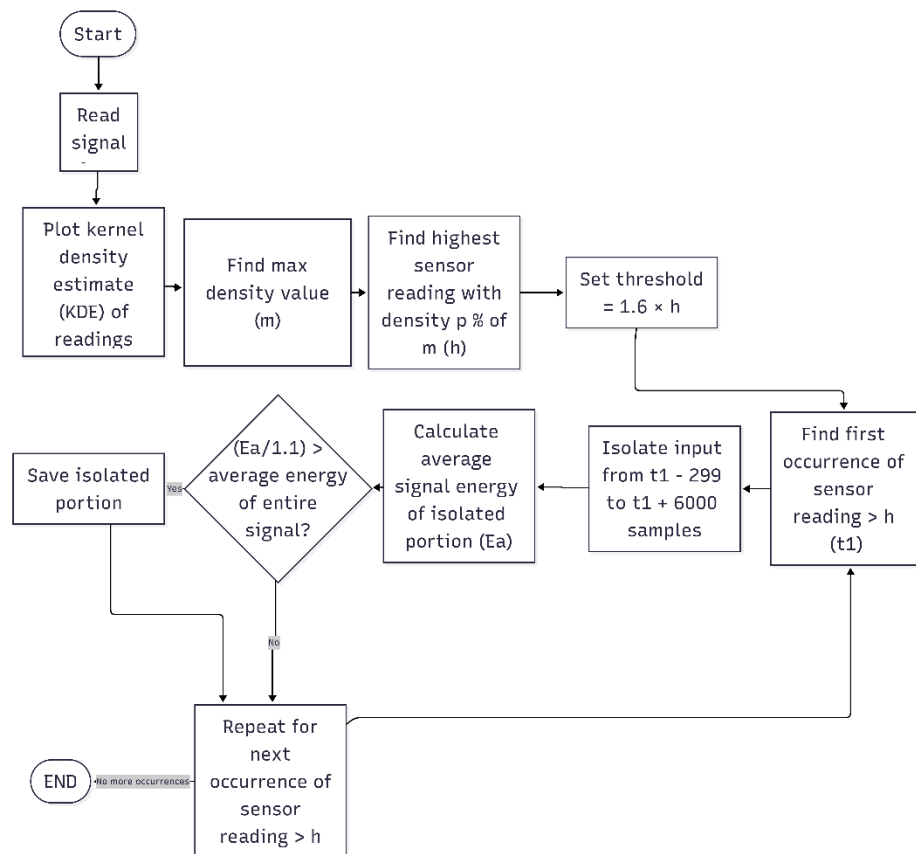


Figure 3.3: Process of isolating PD waveforms.

The process starts by reading the sensor output signal, obtaining a graph looking something like Figure 3.2. Next, the kernel density estimate, using a gaussian kernel, of the signal is plotted. This shows how common it is for each value of the y-axis (sensor reading) to be found.

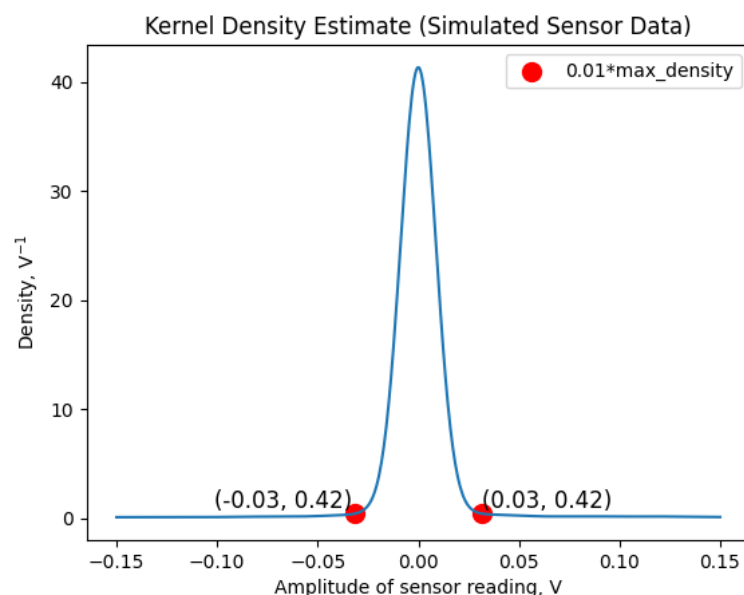


Figure 3.4: KDE plot of the simulated sensor output of Ch1.

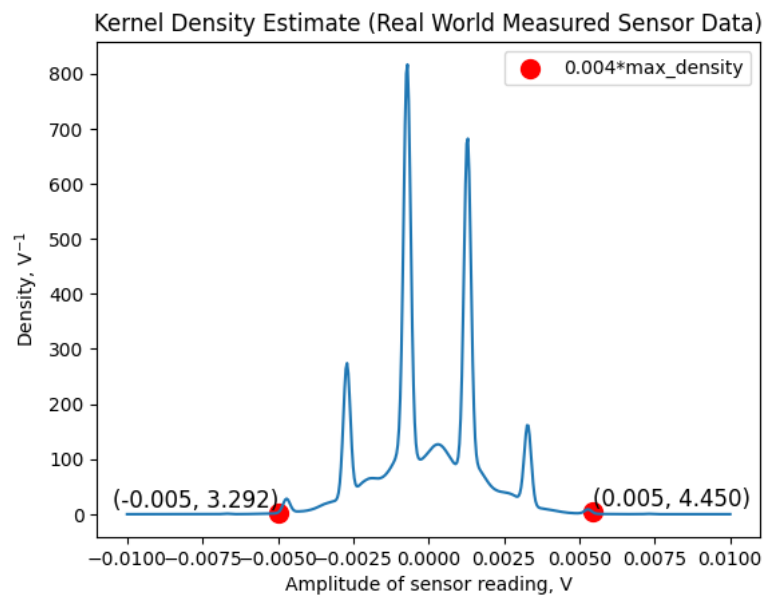


Figure 3.5: KDE plot of a measured sensor output.

As shown in Figure 3.4, values of the sensor reading between -0.03 to 0.03 are most likely to be found (above $p\%$ max density, where p is a value to be configured). In this case, $p = 1\%$. These values represent the noise floor of the signal, while the PD waveforms have values that are more extreme, and occur less frequently. Thus, the maximum value of sensor reading within the $p\%$ max density criterion, multiplied by 1.6 ($0.03 \times 1.6 = 0.048\text{ V}$) is taken as the threshold for first peak detection. Figure 3.5 shows a similar plot, but for real-world measured sensor data. Further details of signal measurement methodology will be discussed in Section 3.5. Note that real world measured noise does not follow a typical gaussian distribution.

The occurrence of a PD event from a sensor's output is then determined by looking at when its value exceeds the set threshold (the first peak occurrence).

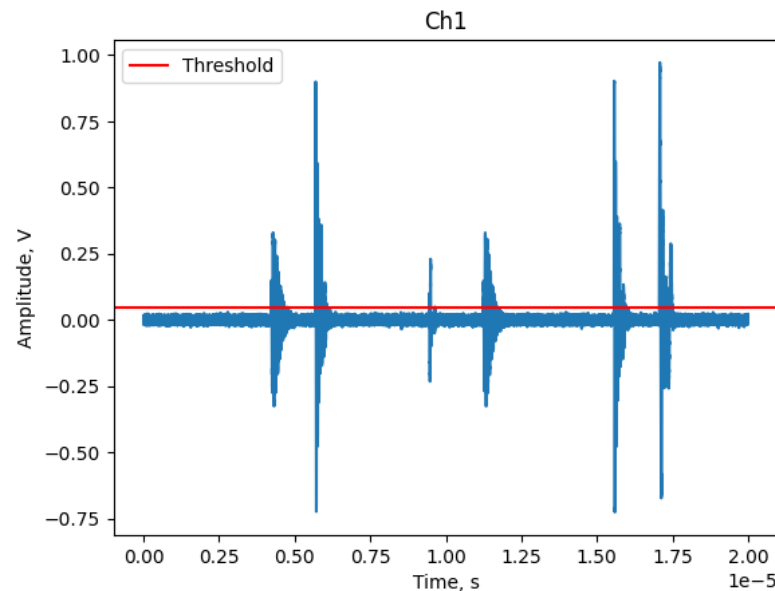


Figure 3.6: Ch1 simulated sensor output, shown with the threshold value for first peak detection.

As shown in Figure 3.6, the threshold accurately separates the noise floor level and the PD signal level. The triggering time is the first time the signal exceeds the threshold. What remains is to isolate the portion of the signal that contains the PD pulse. This is chosen to be between 299 samples before the threshold triggering time, and 6000 samples after it. A final criterion to prevent false triggering is to check whether the average energy of the isolated portion is greater than 1.1 times the average energy of the entire sensor output signal. A PD signal will fulfill this requirement, while an anomalous, unnaturally large and short-lived noise signal would not. The algorithm is looped to process all time indices for which the threshold is exceeded.

3.3 Feature Extraction Methods

Four methods to extract features from the simulated/experimental PD UHF waveform have been identified: wavelet analysis + principal component analysis (Yang & Judd, 2003), envelope detection (Pinpart & Judd, 2010), selected bispectra (He et al., 2020), and statistical analysis (skewness, kurtosis, crest factor, etc).

3.3.1 Wavelet Decomposition (WD) + Principal Component Analysis (PCA)

This combination of techniques was introduced by Hao et al. (2011). The procedure of decomposing the signal using wavelets is called discrete wavelet transform (DWT). DWT is a process in which the signal is broken down into detail and approximate coefficients. Given the original discrete signal $f[x]$, it can be passed through a high pass filter and a low pass filter. The outputs of the filters represent the detail and approximate coefficients respectively, each having half the original signal bandwidth and half the original number of samples. The approximate coefficients can be passed through a subsequent set of filters that further divide the bandwidth and sampling rate. This can be shown in Figure 3.7.

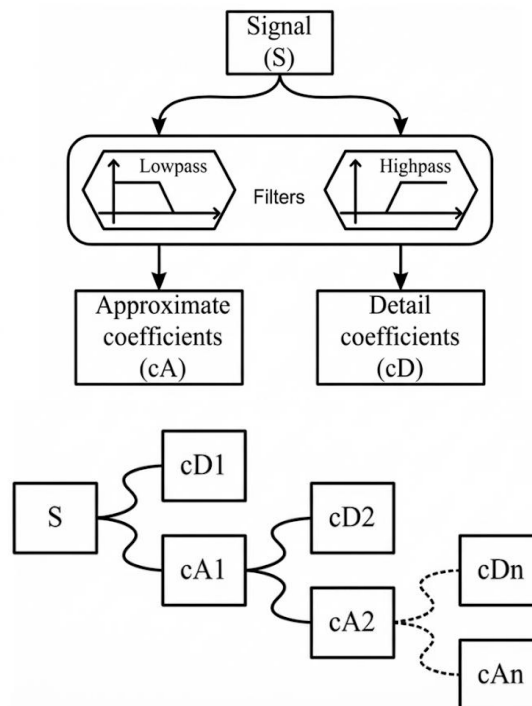


Figure 3.7: Diagrammatic explanation of DWT (Hao et al., 2011).

Mathematically, this can be expressed as:

$$f[x] = \frac{1}{\sqrt{N}} \left(T_{\phi}[0,0]\phi(x) + \sum_{j=0}^{J-1} \sum_{k=0}^{2^j-1} T_{\psi}[j,k]\psi_{j,k}(x) \right) \quad (6)$$

$$T_{\phi}[0,0] = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \phi(x) * f[x] \quad (7)$$

$$T_{\psi}[j,k] = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \psi_{j,k}(x) * f[x] \quad (8)$$

$N = 2^J$ is the number of samples in $f[x]$. $\psi(x)$ and $\phi(x)$ are wavelet and scaling functions, which are the high pass and low pass filters respectively. T_{ϕ} and T_{ψ} are the approximate and detail coefficients. These coefficients are found by taking the inner product of the high pass or low pass filters with the signal $f[x]$, as shown in Equations (7) and (8).

$\psi_{j,k}(x)$ represents the scaled and time shifted version of the original mother wavelet function.

$$\psi_{j,k}(x) = 2^{\frac{j}{2}} \psi(2^j x - k) \quad (9)$$

Where the scaling and time shifting is done on a dyadic grid with j and k being integers starting from 0. At higher values of j , the wavelet function becomes more compact in the time domain, increasing the time resolution while decreasing frequency resolution. Taking the inner product of the wavelet of higher j value with $f[x]$ would cause $T_{\psi}[j,k]$ to show the amplitudes of a wider range of frequencies, and since time resolution increases, the number of samples also increases. $T_{\psi}[j,k]$ is analogous to cDl in Figure 3.7, where $l = n - j, j = 0, \dots, n - 1$.

The scaling function $\phi(x) = \phi_{0,0}(x)$ represents the low pass filter counterpart of $\psi_{0,0}(x)$, producing the final approximate coefficient vector $T_{\phi}[0,0]$, analogous to cAn in Figure 3.7.

An important property that links the mathematical expressions to the recursive filtering process in Figure 3.7, is the expression below.

$$T_{\phi}[j, k] = \sum_n h_{\phi}[n - 2k]T_{\phi}[j + 1, n],$$

$$T_{\psi}[j, k] = \sum_n h_{\psi}[n - 2k]T_{\phi}[j + 1, n]$$
(10)

These show that the detail and approximation coefficients of the next decomposition level can be found by correlating the approximate coefficients of the current level with a signal h_{ψ} or h_{ϕ} (equivalent to convoluting the current level coefficients with the time-reversed version of h_{ψ} or h_{ϕ}), and downsampling by 2. h_{ψ} and h_{ϕ} are sequences of values specific to the type of wavelet and scaling function used.

The wavelet function chosen for the decomposition process is the 9th order ‘Symlet’, and the signal is broken down up to the 5th level, producing cD1, cD2, cD3, cD4, cD5 and cA5. From these decomposition coefficients, their relative signal energies are calculated.

$$\text{Relative energy} = \frac{\text{sum}(\text{square of one set of coefficient})}{\text{sum}(\text{square of all coefficient sets})}$$
(11)

Each signal that goes through the WD process would produce a six-dimensional output of energies.

Principal component analysis (PCA) would then be used to reduce it to three dimensions. PCA is an algorithm used to define orthonormal basis vectors in the directions of highest variance (highest importance), within the dataset, essentially transforming the ‘coordinate system’ of the dataset to a new one. The new coordinate axes that have the highest variance of the data along them are then selected, up to a chosen number. These can represent most of the variance of the dataset and are called the ‘principal components.’ Mathematically,

Let $\mathbf{X}$ be a matrix of six columns, each column being a ‘dimension’ in the data space. This corresponds to the six computed relative energy values from WD. The number of rows are the number of experiments, or in this case, the number of waveforms analyzed.

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_n \end{bmatrix} \quad (12)$$

Where $\mathbf{x}_i, i = 1, \dots, n$ are row vectors, corresponding to the results of each experiment.

The mean of each column is then subtracted from the values, to center each column around zero.

$$\mu_i = \frac{1}{n} \sum_{i=1}^n x_i \quad (13)$$

$$\mathbf{B} = \mathbf{X} - \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \mu_i \quad (14)$$

Then, the covariance matrix of $\mathbf{B}$ is taken

$$\mathbf{C} = \frac{1}{n-1} \mathbf{B}^T \mathbf{B} \quad (15)$$

The eigenvalues and eigenvectors of $\mathbf{C}$ are then computed

$$\mathbf{C}\mathbf{V} = \mathbf{V}\mathbf{D} \quad (16)$$

Where $\mathbf{V}$ is the matrix of eigenvectors, and $\mathbf{D}$ consists of the eigenvalues.

The principal components, $\mathbf{T}$, can then be obtained from:

$$\mathbf{T} = \mathbf{B}\mathbf{V} \quad (17)$$

Intuitively, each column in $\mathbf{V}$ represent the ‘loadings’, or the weighting coefficients of each original axis in the six-dimensional space that make the vector of the new axis (new orthonormal basis) in the transformed coordinate system.

3.3.2 Envelope Detection

The envelope detection method is presented in Pinpart and Judd (2010). The procedure was taken and adapted as follows:

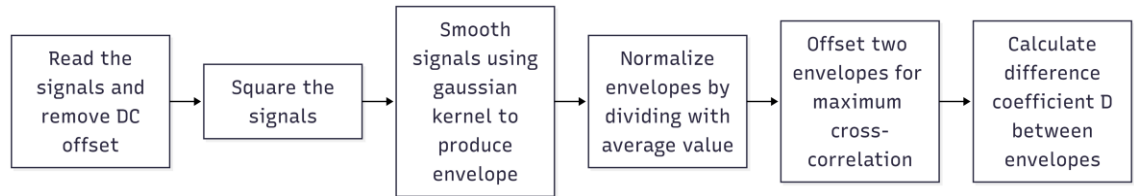


Figure 3.8: Algorithm for creating envelopes and analyzing.

The difference coefficient is defined as.

$$D = \sum_{i=0}^{N-1} (\Delta EN_i)^2 \quad (18)$$

Where ΔEN_i is the difference between two normalized envelopes, at the i^{th} sample point. N is the total number of samples (Pinpart & Judd, 2010).

The purpose of obtaining the envelope is to minimize noise and reduce the sampling rate. Normalization with signal energy instead of max amplitude (what the original source did) is done because some real-world signals have unnaturally high peaks, causing normalized envelopes to be abnormally small in value, increasing D.

3.3.3 Selected Bispectra

The bispectrum of a signal is defined as the two-dimensional Fourier transform of the third-order cumulants of a signal.

Third order cumulants are defined as the expected value of three time-shifted points of a signal $x[n]$,

$$C_{3x}(\tau, \rho) = E[x[n]x[n + \tau_1]x[n + \tau_2]] \quad (19)$$

And consequently, the bispectrum:

$$B(\omega_1, \omega_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} C_{3x}(\tau, \rho) e^{-j(\omega_1 \tau_1 + \omega_2 \tau_2)} d\tau_1 d\tau_2 \quad (20)$$

To estimate the discretized bispectra, the signal is first split into K sections with M observed samples in each section. These are denoted as $x^{(i)}(1), x^{(i)}(2), \dots, x^{(i)}(M)$, where $i = 1, 2, \dots, K$. Then, the Fourier transform coefficients of each section are obtained.

$$X^{(i)}(\lambda) = \frac{1}{M} \sum_{n=1}^M x^{(i)}[n] e^{-j\left(\frac{2\pi(n-1)\lambda}{M}\right)} \quad (21)$$

And $\lambda = 0, 1, \dots, M/2$. The bispectra of each section of the signal is

$$\begin{aligned} & \tilde{b}^{(i)}(\lambda_1, \lambda_2) \\ &= \frac{1}{\Delta^2} \sum_{i_1=-L}^L \sum_{i_2=-L}^L X^{(i)}(\lambda_1 + i_1) X^{(i)}(\lambda_2 + i_2) X^{(i)*}(-\lambda_1 - \lambda_2 - i_1 - i_2) \end{aligned} \quad (22)$$

With $\Delta = f_s/N_0$, and $M = (2L + 1)/N_0$. Finally, the bispectra estimation of the whole signal is:

$$\tilde{B}(\lambda_1, \lambda_2) = \frac{1}{K} \sum_{i=1}^K \tilde{b}^{(i)}(\lambda_1, \lambda_2) \quad (23)$$

This method is presented by He et al. (2020), because the bispectra is insensitive to Gaussian noise, leaving only the non-Gaussian PD signal to be visible.

3.3.4 Statistical Analysis

This would be the simplest feature extraction method of the four. Simple calculations are performed on the PD waveform, extracting features shown below (Carvalho et al., 2024):

Peak value (V_p):

$$V_p(x) = \max(x_i), i = 1, 2, \dots, N \quad (24)$$

Root mean square value (V_{RMS}):

$$V_{RMS}(x) = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2} \quad (25)$$

Average amplitude (A_a):

$$A_a(x) = \frac{1}{N} \sum_{i=1}^N |x_i| \quad (26)$$

Square root amplitude (A_{r2}):

$$A_{r2}(x) = \left(\frac{1}{N} \sum_{i=1}^N \sqrt{|x_i|} \right)^2 \quad (27)$$

Crest factor (C_f):

$$C_f(x) = \frac{V_p(x)}{V_{RMS}(x)} \quad (28)$$

Clearance factor (CL_f):

$$CL_f(x) = \frac{V_p(x)}{A_{r2}(x)} \quad (29)$$

Impulse factor (I_f):

$$I_f(x) = \frac{V_p(x)}{A_a(x)} \quad (30)$$

Shape factor (S_f):

$$S_f(x) = \frac{V_{RMS}(x)}{A_a(x)} \quad (31)$$

Skewness (S_k):

$$S_k(x) = \frac{E(x - \mu)^3}{\sigma^3} \quad (32)$$

Kurtosis (K_u):

$$K_u(x) = \frac{E(x - \mu)^4}{\sigma^4} \quad (33)$$

These features would then be input into other algorithms, such as a deep learning model, Gaussian mixture model or K-means clustering algorithm for clustering.

3.4 Clustering/Separation Methods

Several different clustering/separation methods have been identified during the literature review, such as the K-means algorithm, Gaussian mixture model, density based spatial clustering of applications with noise (DBSCAN), and deep learning (DL).

3.4.1 K-means Clustering

This is the most classical clustering algorithm. Beforehand, the number of clusters, K , must be defined. Then, given a dataset with a certain number of readings (points), each having a certain number of features (the dimensions of the dataset),

- iv. Randomly select K cluster centers.
- v. Assign each point in the dataset to the nearest cluster center.
- vi. Calculate the mean of each cluster and take that as the new cluster centers.
- vii. Repeat step 2 and 3 until the change of the cluster center from one iteration to the next is less than a set threshold.

K-means clustering is used to cluster PD information in (Carvalho et al., 2024) and (Zhou et al., 2022).

3.4.2 Gaussian Mixture Model (GMM)

This model assumes that the data set consists of several normally distributed clusters. It calculates the probability that a point belongs to the clusters around it.

$$P(z_n = k | x_n) = \frac{\pi_k N(x_n | \mu_k, \Sigma_k)}{\sum_{k=1}^K \pi_k N(x_n | \mu_k, \Sigma_k)} \quad (34)$$

Equation (34) is the probability that a point in the dataset x_n belongs to the Gaussian density cluster k (z_n is the cluster number). $N(x_n | \mu_k, \Sigma_k)$ is the multivariate Gaussian density with unknown mean μ_k and unknown covariance Σ_k . π_k is the unknown probability of selecting the k^{th} cluster (GeeksforGeeks, 2025).

The estimation of the mean and covariance of each cluster is then done by the Expectation-Maximization Algorithm, optimizing the log-likelihood of the mixture model.

3.4.3 Density Based Spatial Clustering of Applications with Noise (DBSCAN)

Unlike K-means and GMM, DBSCAN does not require the pre-determination of the number of clusters. Instead, it requires two user-defined parameters:

- i. ϵ : the radius of the hypersphere around a data point
- ii. min_samples: the minimum number of data points required to form a cluster.

The algorithm classifies each point in the dataset as:

- i. Core point: it has min_samples of other points within a hypersphere of radius ϵ from it.
- ii. Border point: it has fewer than min_samples around it, but it is near a core point.
- iii. Noise point: do not belong to any cluster.

Finally, the DBSCAN algorithm is as follows:

- i. Loop through the dataset and identify all core points.
- ii. Create a new cluster for each core point that is not associated with any cluster.
- iii. Add points within the hypersphere of a core point into its cluster.
- iv. Add density connected points into the clusters. A density connected point is a point for which there exists a chain of connected points from it to a core point, with each point being within ϵ from the next.
- v. After processing all points, any points that aren't clustered become noise points.

3.4.4 Deep Learning (DL)

While traditional machine learning methods rely on manually engineered features—such as the statistical analysis described in Section 3.3.4—Deep Learning (DL) approaches automate the process of feature extraction. DL is a subset of machine learning based on artificial neural networks (ANNs), which are computing systems inspired by the biological neural networks of animal brains. These networks consist of multiple layers of "neurons" that progressively extract higher-level features from the raw input data.

For the analysis of Partial Discharge signals, particularly when transformed into the time-frequency domain, Convolutional Neural Networks (CNNs) are the most prominent architecture. A CNN is designed to process data that has a known grid-like topology, such as time-series data (1D) or image data (2D).

The core operation of a CNN is the "convolution," where a set of learnable filters (or kernels) scans the input data. As the filter slides across the input, it performs element-wise multiplication and summation, producing a "feature map."

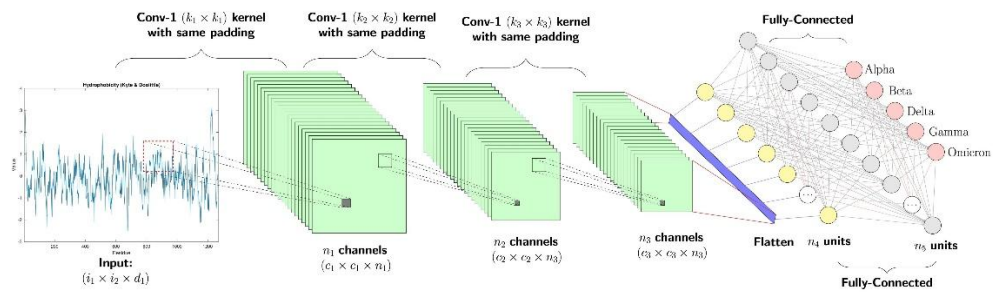


Figure 3.9: A brief overview of how a CNN works (Priyanshinavadiya, 2025).

This process allows the network to detect local patterns—such as the sharp rise times or specific oscillatory decay of a PD pulse—regardless of their position in the time window. As the data passes through deeper layers of the network, these local patterns are combined to form complex, high-level representations of the signal structure.

In this study, a CNN will eventually be constructed to cluster PD signals using open-source tools like PyTorch and Tensorflow.

3.5 Experimental Measurement

An experiment was conducted to obtain a set of real-world measured PD waveforms, to test the capabilities of the clustering methods. It was conducted in an indoor environment, in a high voltage cage. Two different types of physical PD models were used. The first was a void defect in epoxy resin insulation, mimicking internal gas pockets, with a 1.5mm energized copper wire and 0.5mm grounded copper strip. The second was similar to the first, but the defect was in the form of an incision, mimicking mechanical damage to electrical insulation. The models were immersed in transformer oil to suppress surface discharge. Next, the electrodes on the model were energized to AC voltages of 8kV for the void defects, 17kV for the incision, and 15kV when both models were simultaneously tested. The complete design details of the models are specified in a prior work by Madhuranga et al. (2025).

Four UHF sensors were used to collect PD EM wave emissions. The sensors were planar monopole antennae (PMA) developed by Kayser Azam et al. (2024). Each sensor

was connected to a four-channel digital oscilloscope via coaxial cables, operating at 10GS/s. According to the Nyquist-Shannon sampling criterion, this allows continuous-time signals of up to 5GHz to be reconstructed from the sampled data.

The experimental setup is given in Figure 3.10. The sensor coordinates are chosen to improve accuracy and coverage area of PD localization, based on preliminary testing done by Chan et al. (2026).

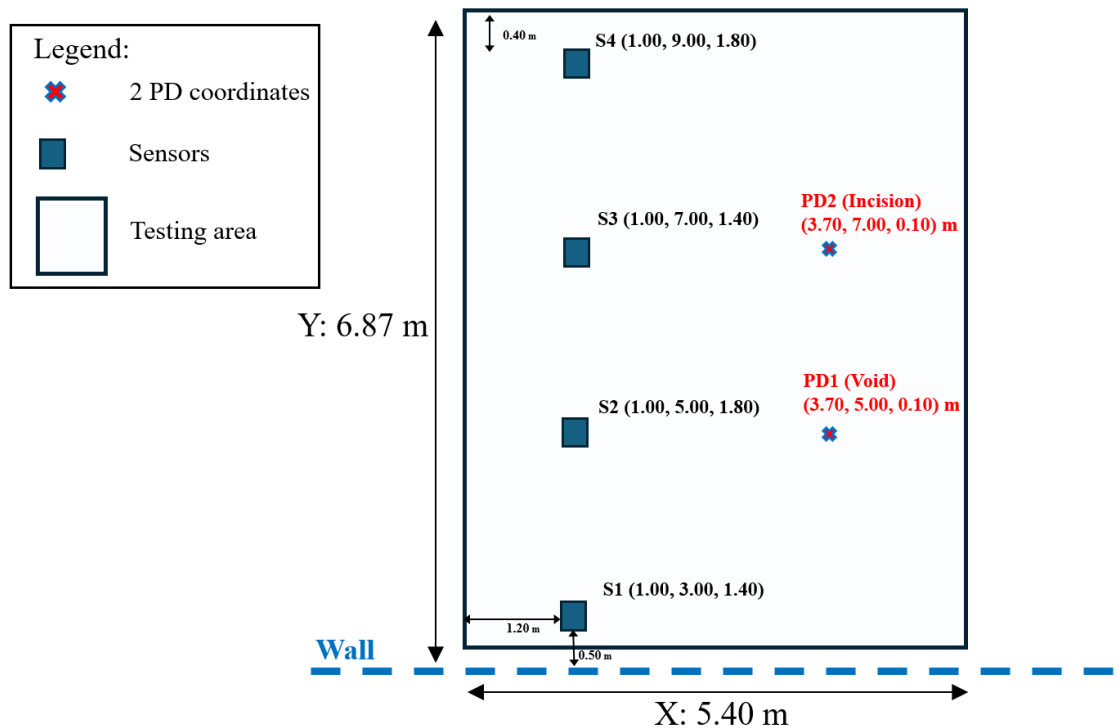


Figure 3.10: Schematic diagram of experimental layout.

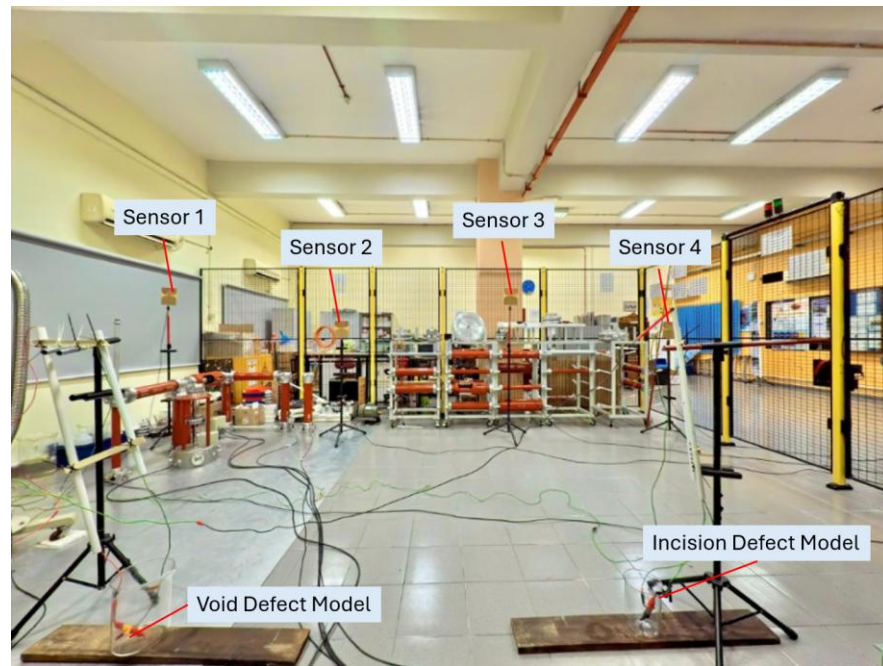


Figure 3.11: Actual experimental layout.

When the PD model is energized, a threshold is set on the oscilloscope to capture the instant in time where the reading exceeds it (which is when a PD waveform is observed). The readings of all four channels are then saved as one set. In total, three sets of waveforms are recorded for each defect type. In addition, when both PD models were energized, oscilloscope readings were chosen such that two PD waveforms could be observed in each channel reading. Two sets of these were also recorded. In total, there were 40 recorded signals from all channels and reading sets. However, only 24 were used in the analysis, which were the ones in which only a single PD was energized, so that the PD type for each signal is known.

3.6 TDOA Estimation Method

TDOA is estimated using the cross-correlation technique. This involves a very similar process to Envelope Analysis, shown in Figure 3.8, but for the final step, instead of obtaining the difference coefficient, the offset for maximum cross-correlation is taken and used to calculate TDOA by dividing by sampling frequency, f_s .

3.7 Localization Technique

The particle swarm optimization (PSO) technique was used to produce results of localization, for comparison purposes. Since the focus of this research is not to optimize PD localization of one PD source, only one technique will be employed.

3.8 General Clustering and Localization Algorithm

For data clustering, a few different combinations of DBSCAN, PCA, WD, and Envelope Analysis were experimented on. Two different stages of the algorithm are defined. Stage 1 of the process would be as in Figure 3.12.

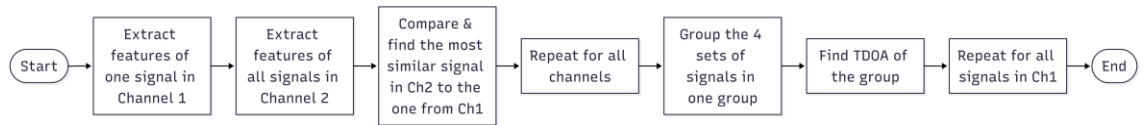


Figure 3.12: Algorithm for Stage 1.

After Stage 1 is performed, and the signals are divided into their groups, Stage 2 begins. The algorithm assumes that the 4 sets of signals in a group come from the same PD source, emitted at the same instant. PSO and further clustering are performed to obtain the final estimate of PD locations.

3.9 Quantitative Results Analyses

Stage 1's accuracy is measured by how many groups of signals it correctly clustered. A correct group of signals is whether each signal from the group is from the same PD source, emitted at the same instant.

$$\text{Stage 1 Accuracy} = \frac{\text{Number of correct groups}}{\text{Total number of groups formed}} \times 100\% \quad (35)$$

Stage 2's accuracy can be measured in two ways:

$$\text{PD Number Accuracy Ratio} = \frac{\text{Number of PDs obtained}}{\text{Actual number of PDs}} \quad (36)$$

$$E = \text{Total euclidean distance from all estimated PD to actual PD} \quad (37)$$

$$\text{Avg. Euclidean Distance} = \frac{E}{\text{Estimated number of PDs}}$$

The average Euclidean distance parameter is calculated by first taking the distances of all estimated PD locations to their actual PD locations. However, if the estimated PD location is wrong (for example, if excessive error in Stage 1 caused Stage 2 to localize for the wrong signal sets), the closest actual PD location is taken.

CHAPTER 4: PRELIMINARY RESULTS AND DISCUSSION

4.1 Measured PD Waveforms

The measured PD waveforms are shown in Figure 4.1 and Figure 4.2. It can be noted that the shape of the measured waveforms resembles the second and fourth simulated waveforms: a SEDO with uneven oscillations near the tail end.

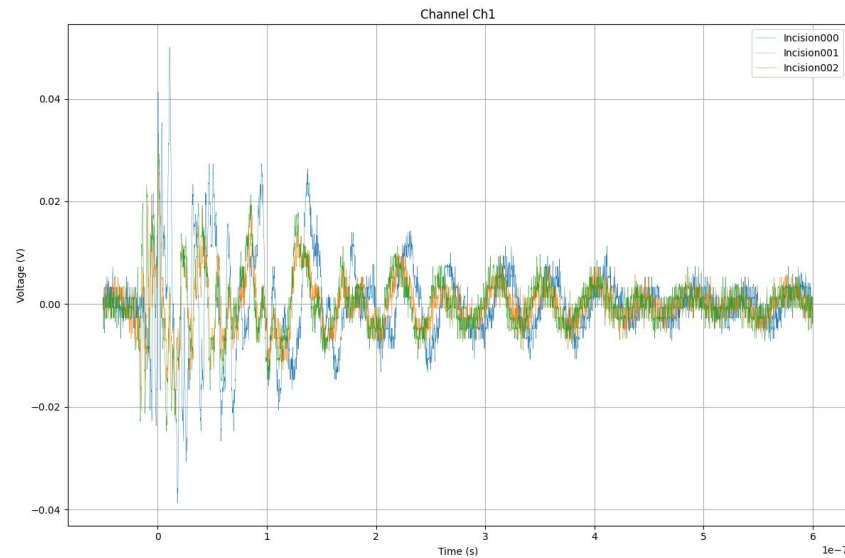


Figure 4.1: Measured PD waveforms of the incision defect on sensor 1 (Ch1).

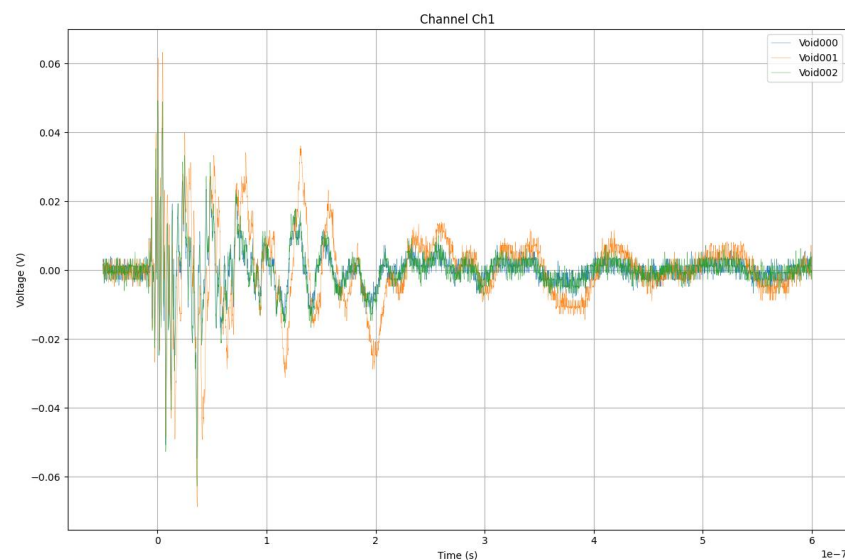


Figure 4.2: Measured PD waveforms of the void defect on sensor 1 (Ch1).

4.2 Analysis on Simulated Waveforms

4.2.1 Technique 1: Envelope Analysis, PSO and then DBSCAN

The four channels of simulated waveforms were put through one round of envelope analysis. For Stage 1, the signals with the lowest difference coefficient value between each other, across the four channels, were chosen to be in one group (representing one PD from one time instance), according to the procedure in Figure 3.12. Then, for Stage 2, the TDOA was calculated for each group. The coordinates of the PD were then estimated for each group using PSO, and DBSCAN ($\epsilon = 1, \text{min_samples} = 2$) was used to cluster the resulting PD coordinates. The mean of the coordinates in each group was taken to be the final estimated PD location.

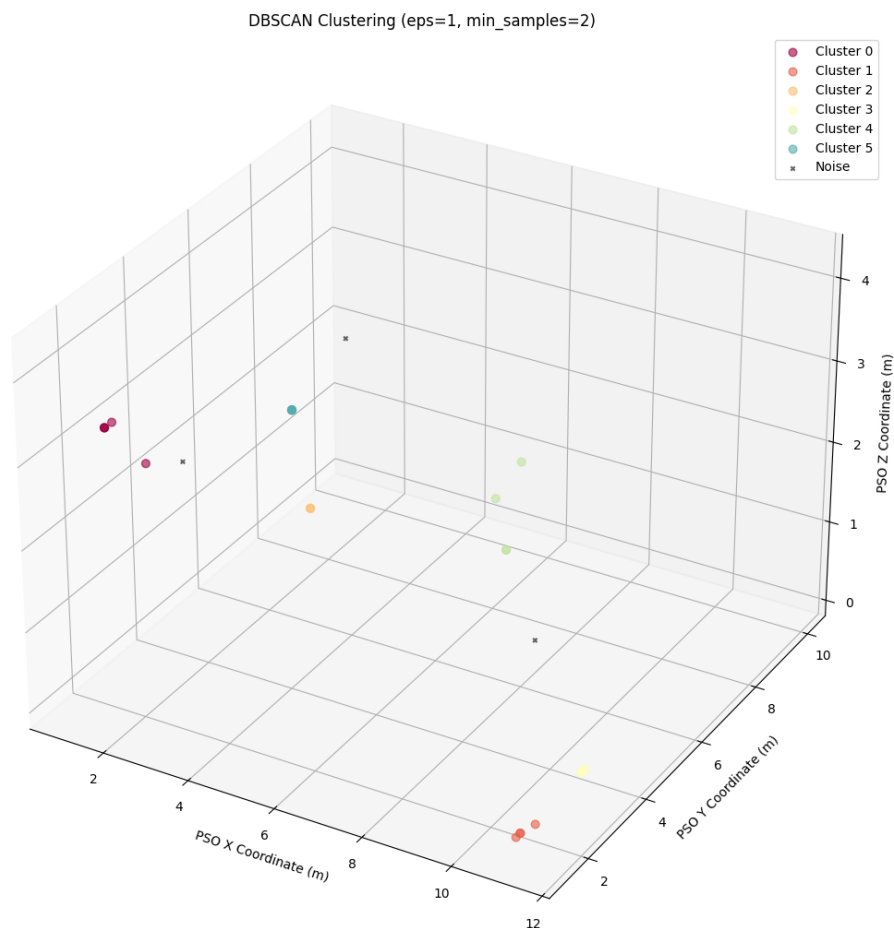


Figure 4.3: DBSCAN clustering of PSO coordinates.

Table 4.1 : Stage 2 results for simulated waveforms, Technique 1.

cluster (PSO -> DBSCAN)	Est. X (m)	Est. Y (m)	Est. Z (m)	Actual X (m)	Actual Y (m)	Actual Z (m)	Euclidean Distance (m)
0	1.2560	2.284000	3.066000	1.2	2.3	3.1	0.06743886120034963
1	11.1000	1.407500	0.155000	2.9	6.3	5.4	10.894337118429924
2	6.1500	1.760000	3.090000	6.1	1.7	3.5	0.4173727350941844
3	11.0000	3.575000	0.100000	4.5	7.8	6.2	9.864614792276484
4	9.3325	3.475000	2.952500	9.2	3.5	2.8	0.20356202985822325
5	5.8500	1.650000	4.220000	7.8	1.2	4.9	2.113622482847872

Table 4.2: Quantitative results analyses for simulated waveforms, Technique 1.

Stage 1 Accuracy	96.77% (30 out of 31)
PD Number Accuracy Ratio	6/6
Avg. Euclidean Distance	3.9268247 m

4.2.2 Technique 2: Envelope Analysis, DBSCAN, and then PSO

In this technique, the same Stage 1 results of Technique 1 were put through a reverse order of Stage 2, i.e.: DBSCAN on the TDOAs, taking the mean of the TDOAs in each group, and then localizing with PSO using the mean TDOAs.

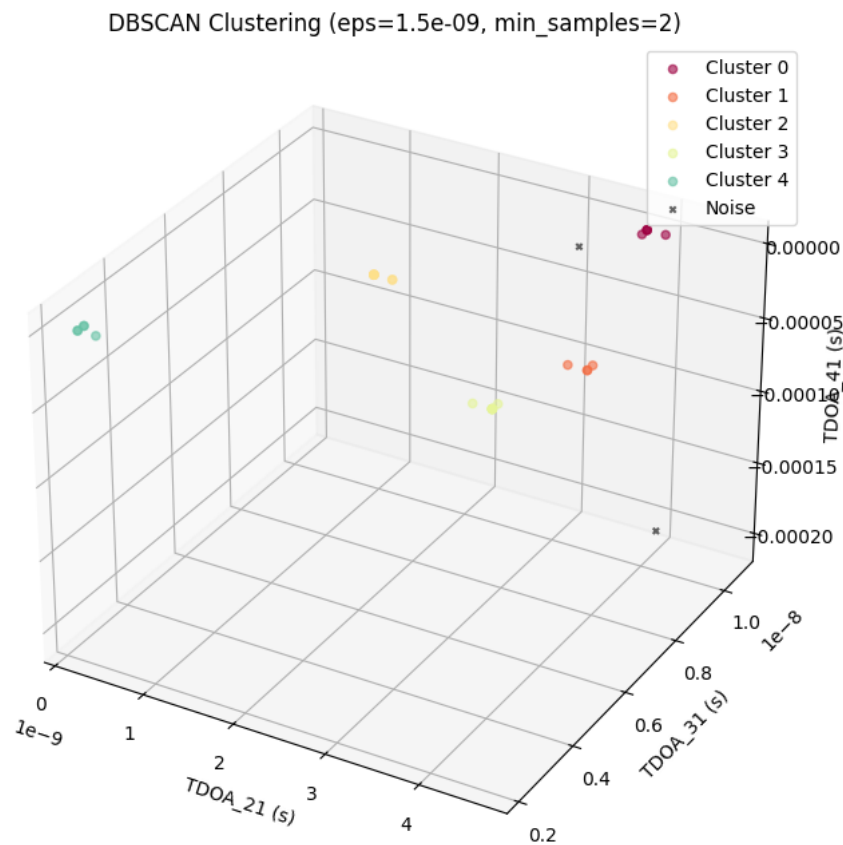


Figure 4.4: DBSCAN clustering of TDOA values.

Since DBSCAN was performed on TDOA values instead of coordinates, a lower $\epsilon = 1.5 \times 10^{-9}$ value is used.

Table 4.3: Stage 2 results for simulated waveforms, Technique 2.

cluster (DBSCAN- > PSO)	Est. X (m)	Est. Y (m)	Est. Z (m)	Actual X (m)	Actual Y (m)	Actual Z (m)	Euclidean Distance (m)
0	1.08	2.26	3.14	1.2	2.3	3.1	0.1326649916142159
1	11.0	1.43	0.1	2.9	6.3	5.4	10.83590789920254
2	5.65	2.34	1.5	6.1	1.7	3.5	2.14758003343298
3	11.0	3.52	1.04	7.8	1.2	4.9	5.524671935961448
4	9.07	3.64	1.68	9.2	3.5	2.8	1.1361778029868386

Table 4.4: Quantitative results analyses for simulated waveforms, Technique 2

Stage 1 Accuracy	96.77% (30 out of 31)
PD Number Accuracy Ratio	5/6
Avg. Euclidean distance	3.95540 m

4.2.3 Technique 3: WD+PCA, PSO, and then DBSCAN

For this technique, Stage 1 involves performing WD and PCA to extract the 3 principal components of each signal. Then, grouping is done based on which signal in channels 2, 3 and 4 are nearest to a particular channel 1 signal. This is done by using the 3 principal components as coordinates and calculating the Euclidean distance between signals.

Stage 2 is the same as Technique 1.

Table 4.5: Quantitative results analyses for simulated waveforms, Technique 3

Stage 1 Accuracy	12.90% (4 out of 31)
PD Number Accuracy Ratio	N/A
Avg. Euclidean distance	N/A

It should be noted that only 4 signal groups were clustered correctly in Stage 1. During PSO and subsequently DBSCAN, two of these groups were clustered as noise, and 2 of them were clustered under cluster 4.

Therefore, the only meaningful result would be the Euclidean distance between cluster 4 and the nearest actual PD coordinates.

Table 4.6: Stage 2 results for simulated waveforms, Technique 3.

cluster (DBSCAN- > PSO)	Est. X (m)	Est. Y (m)	Est. Z (m)	Actual X (m)	Actual Y (m)	Actual Z (m)	Euclidean Distance (m)
4	11.0	3.57	0.1	9.2	3.5	2.8	3.245751068705054

4.2.4 Technique 4: WD+PCA, DBSCAN, and then PSO

For this technique, Stage 1 for Technique 3 and Stage 2 of Technique 2 are combined. Once again, out of the 4 correct signal groups from Stage 1, only 2 were clustered into Cluster 0.

Table 4.7: Stage 2 results for simulated waveforms, Technique 4.

cluster (DBSCAN- > PSO)	Est. X (m)	Est. Y (m)	Est. Z (m)	Actual X (m)	Actual Y (m)	Actual Z (m)	Euclidean Distance (m)
0	11.0	3.56	0.21	9.2	3.5	2.8	3.1546315157241427

It can be concluded that WD + PCA is not a good representation of the signal. This can be explained by the fact that the underlying PD signal is simulated at low frequencies before adding high-frequency noise on top of it. The farther the distance that the emitted PD signal travels to the receiver, the more it attenuates. This makes the energy of the low frequency component, lower at the receiver. However, noise remains at the same power regardless of where the receiver is located. This leads to different SNRs for PD signals from different origins, detected at a particular receiver, which highly affects the output relative energy levels of the WD process.

WD decomposes the received signal into its component frequencies (in this case, six of them). Higher-level coefficients like cA5 represent the original, underlying low frequency PD signal, while lower-level coefficients cD2 and cD1 represent high frequency noise. Then, the energies of each coefficient are calculated and normalized based on total energy of all coefficients. This is where the problem arises, because regardless of the different attenuation levels of the underlying PD signal, its energy is still orders of magnitude larger than noise, and the **normalized** energy of cA5 will represent more than 99% of the total energy of all coefficients. Before normalization happens, different SNRs are reflected by constant high-frequency noise power but different low-frequency PD signal power. After normalization, relative energy of low-frequency coefficient cA5 is almost constant at 99% and above, regardless of PD signal attenuation levels. Therefore, the SNR variation will then be reflected in varying high-frequency coefficient energies of cD2 and cD1.

After going through PCA, it selects three components of the six-dimensional set of relative energy levels that explain the highest variance. Since cA5 is relatively constant across all received PD signals on all channels, this will not greatly influence the resulting three components. What will greatly influence them are cD2 and cD1, which vary greatly based on how far the PD travels to the receiver. Thus, what influences clustering is not just PD type, but also the distance travelled. This can be visualized in the plots below.

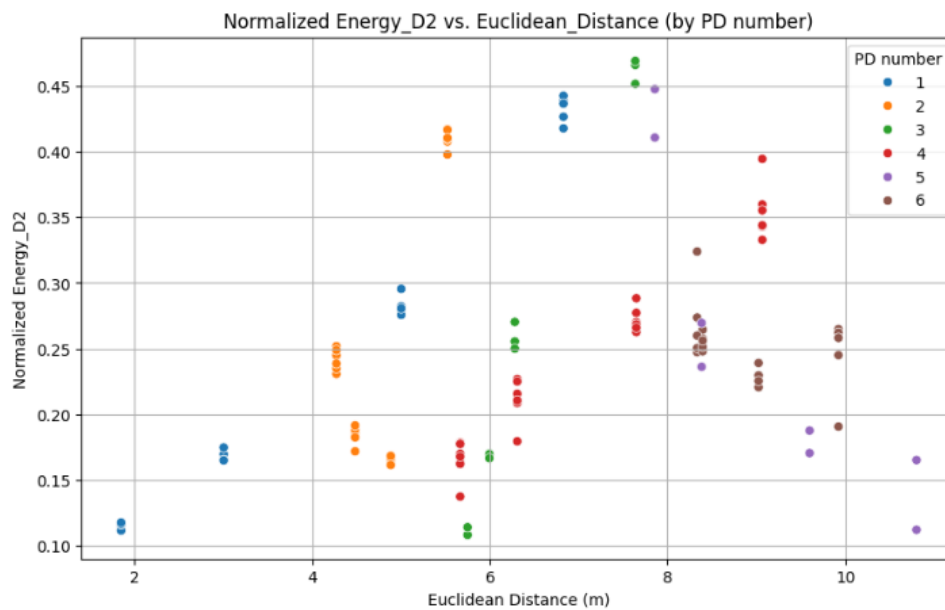


Figure 4.5: Normalized energies of D2 coefficients against distance between PD source and receiver.

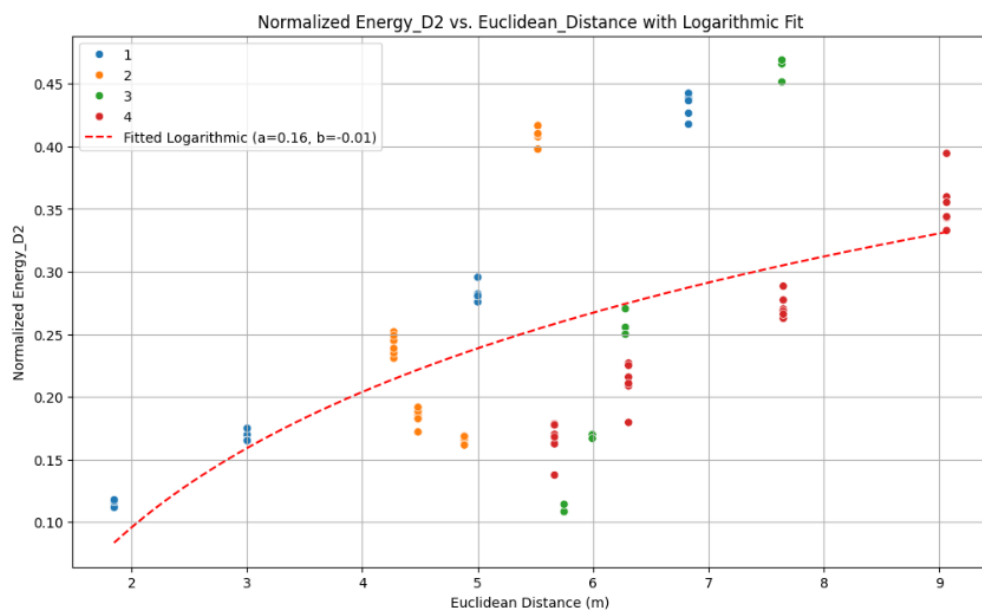


Figure 4.6: Fitted logarithmic function against the values for PD 1-4.

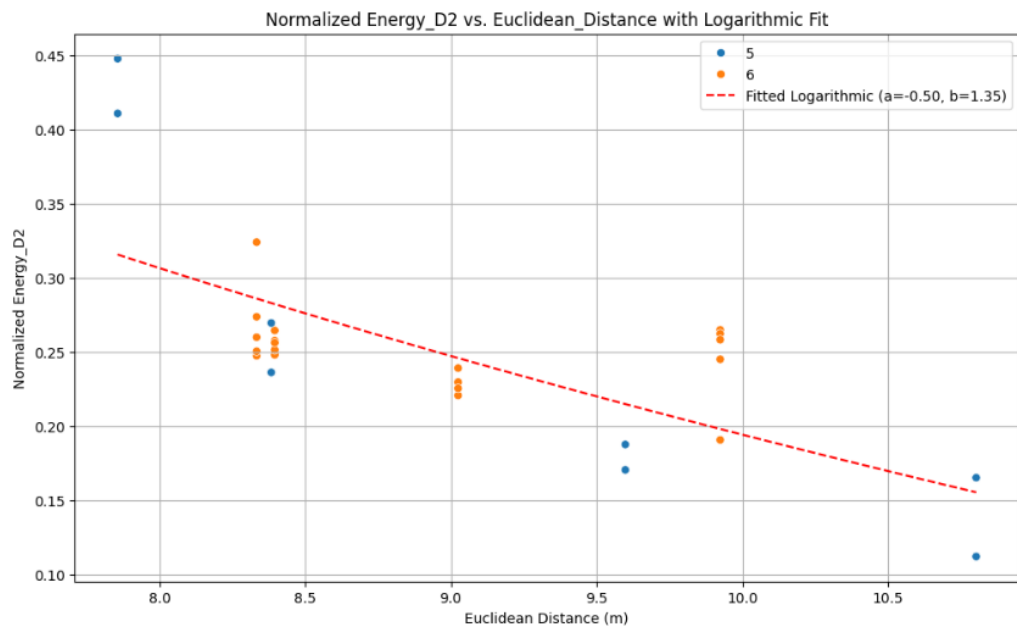


Figure 4.7: Fitted logarithmic function against the values for PD 5-6.

A clear correlation between the distance to the receiver and the received energy of cD2 can be seen in Figure 4.6. Conversely, a reverse pattern is found in Figure 4.7, with PDs 5 and 6. This may be due to their abnormal shape which affects the high frequency coefficient.

4.3 Analysis on Measured Waveforms

Stage 1 for Technique 1 and 2 could not correctly separate measured signals into their correct groups. This is due to how void and incision waveforms are highly similar to each other, in terms of the difference coefficient, D .

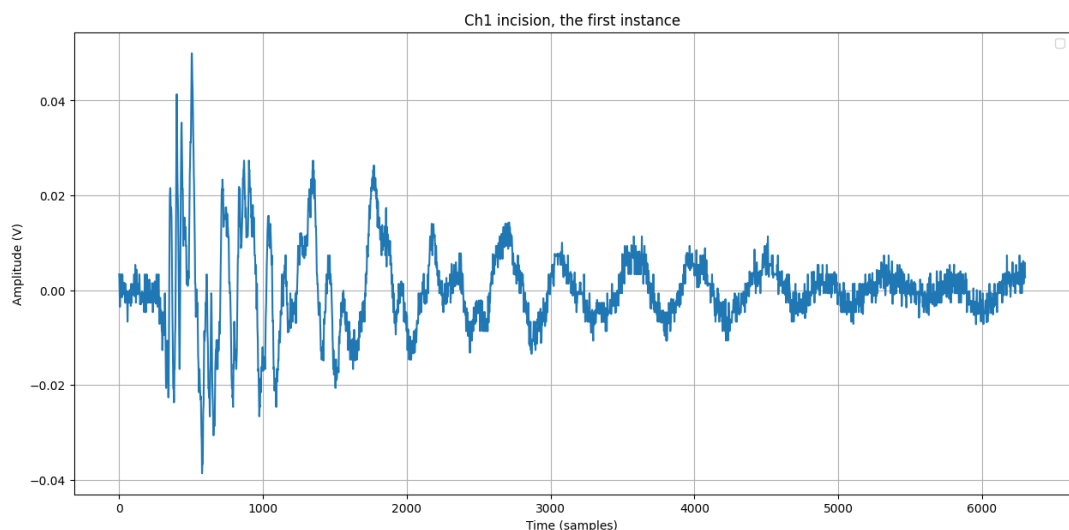


Figure 4.8: The first instance of incision PD discharge of Ch1.

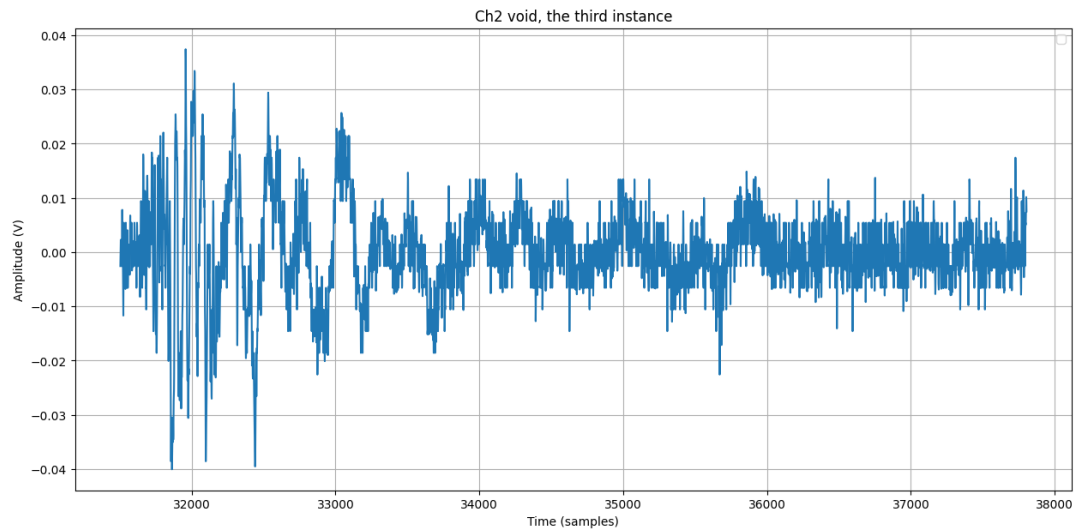


Figure 4.9: The third instance of void PD discharge of Ch2.

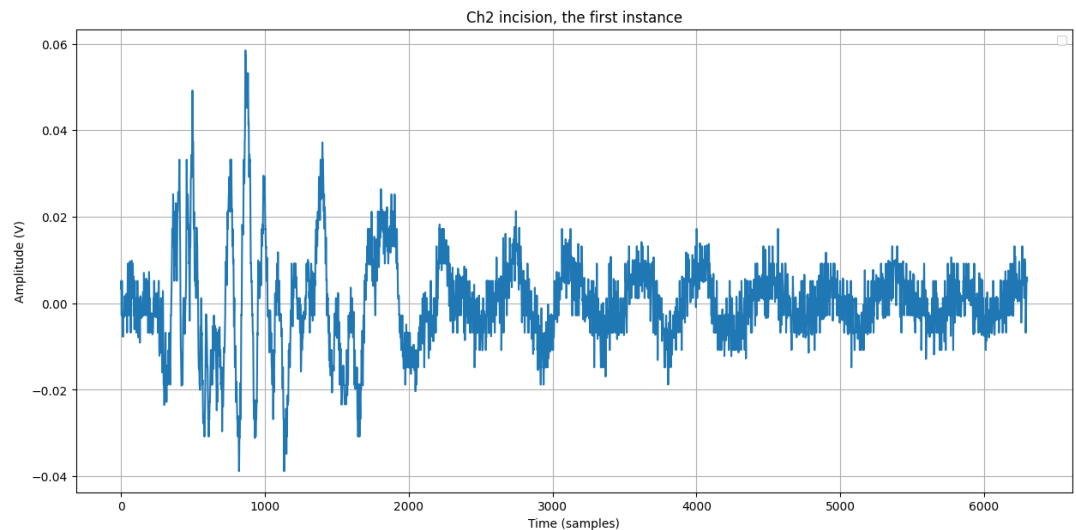


Figure 4.10: The first instance of incision PD discharge of Ch2.

To illustrate how easily the algorithm made the error, Figure 4.8 to Figure 4.10 are shown. The algorithm wrongly matched the first instance of incision PD discharge in Ch1, with the third instance of void PD discharge in Ch2, instead of matching it with the first instance of incision PD discharge in Ch2. The envelopes of both waveforms from Ch2 are highly similar, and the scalar, 1 dimensional difference coefficient, D , was simply not enough to differentiate the two signals well.

For Technique 3 and 4, out of all the six groupings done, the WD + PCA algorithm mistakenly grouped void and incision defects together in four of them. For the remaining two groups, despite the PD type being homogenous for all signals within one group, the

algorithm incorrectly placed PD signals from different emission instances together. Thus, no group was correctly clustered.

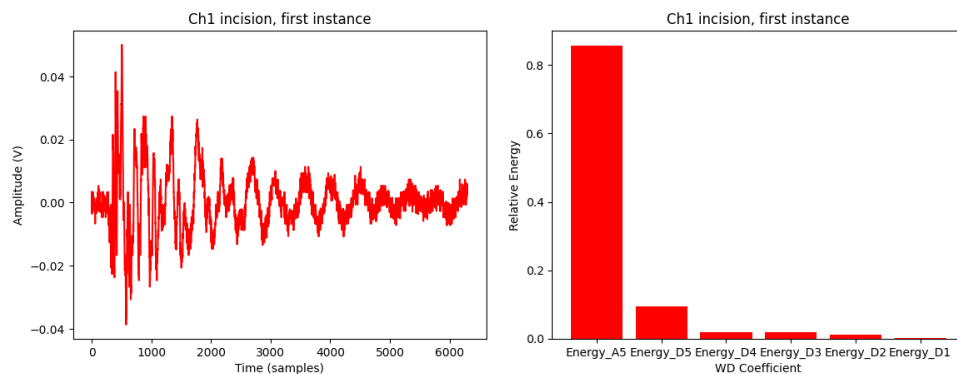


Figure 4.11: Signal and WD energies of Ch1 incision, first instance.

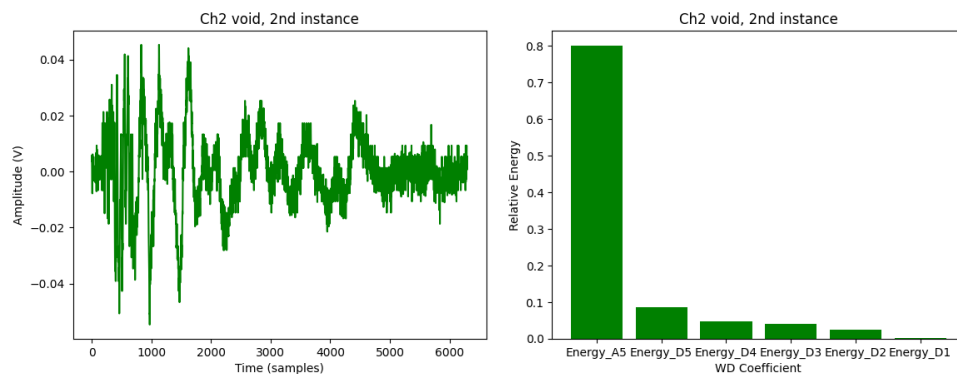


Figure 4.12: Signal and WD energies of Ch2 void, second instance.

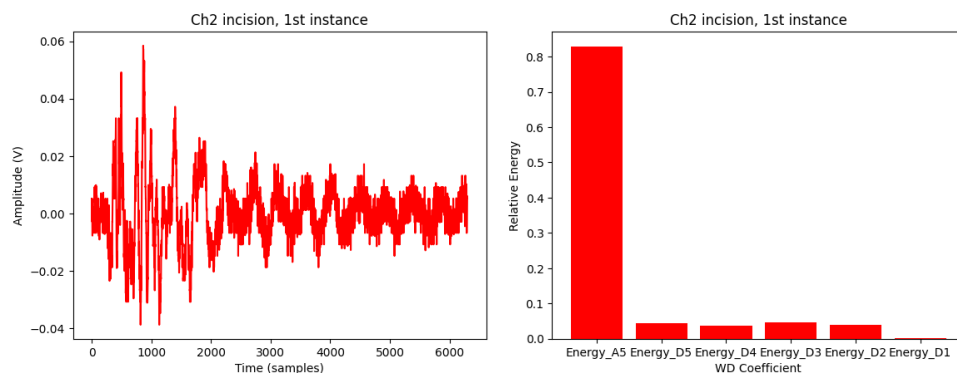


Figure 4.13: Signal and WD energies of Ch2 incision, first instance.

The above figure shows why the WD + PCA algorithm wrongly grouped two signals, as well as the correct signal that should be in that group. The algorithm placed the second void PD instance of Ch2 closer to the first instance of incision PD of Ch1, instead of the first incision instance of Ch2. The shape of the energy content graph of WD shows a slightly more left-skewed shape for the two signals that were wrongly grouped together,

resulting in PCA output coordinates that are closer than what they should be. Thus, it can be said that the energy distribution of WD coefficients does not accurately represent the difference between each signal.

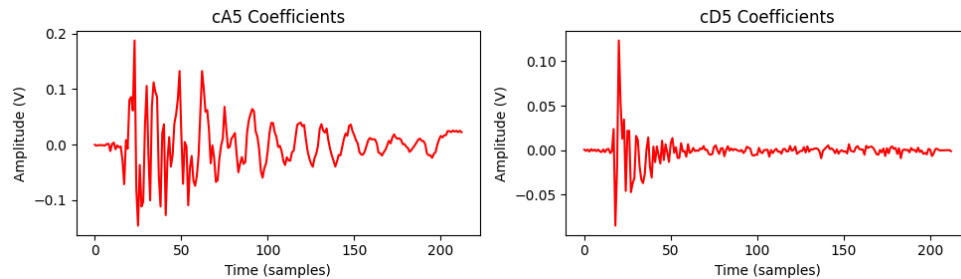


Figure 4.14: WD first two coefficients of Ch1 incision, first instance.

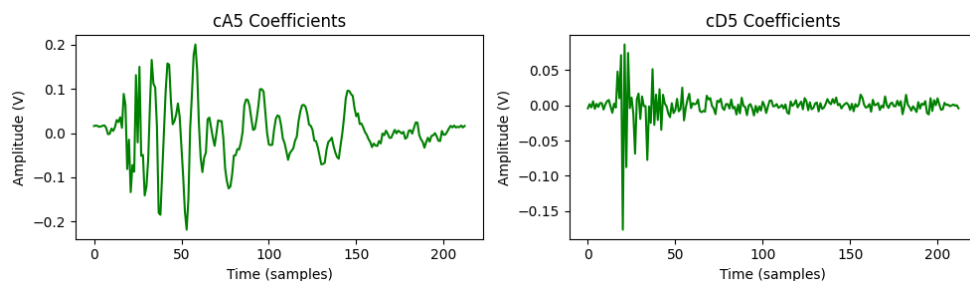


Figure 4.15: WD first two coefficients of Ch2 void, second instance.

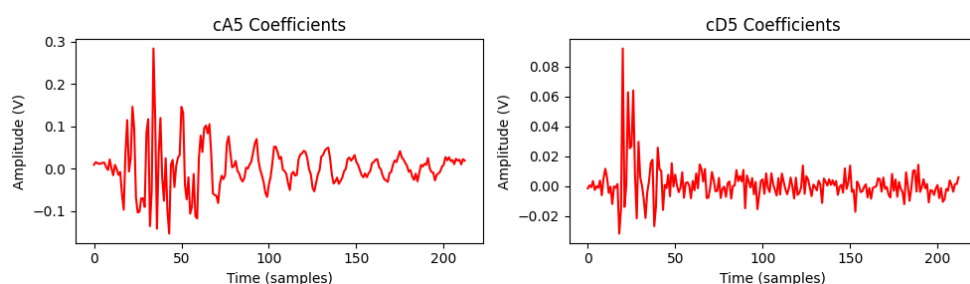


Figure 4.16: WD first two coefficients of Ch2 incision, first instance

It can be seen even more clearly how the second coefficient of Ch2 incision (first instance) has a comparatively lower amplitude than the other two, contributing to its dissimilar energy distribution graph. In addition, the same PD emitted signal undergoes attenuation, the further it has to travel to a receiver. This affects the results for reasons explained in Section 4.2.4. Next, it was observed that the average energies of cD2 and

cD1 (before normalization) for Ch1 were approximately four times less than the others. The reason for this phenomenon is unexplained, but it could be because sensor 1 was located 0.5 m to a wall, meaning the effects of multipath propagation and attenuation of noise was greater at that location. On the other hand, it could also be because of connection errors between sensor 1 and the oscilloscope, or a defect in sensor 1 itself. This will influence the results of PCA that is subsequently performed.

Table 4.8: Average energies of noise coefficients for each channel.

Channel	Average cD2 energy, before normalization	Average cD1 energy, before normalization
1	0.005528035	0.000399873
2	0.027646102	0.001756094
3	0.026863238	0.001755848
4	0.028222984	0.001744885

One more thing to note is that the cA5 coefficients for both incision waveforms were highly similar, and both differed greatly from the cA5 coefficients of the void waveform.

4.4 Discussion and Future Work

From the current preliminary results, it can be concluded that, despite Envelope Analysis being an effective method to cluster simulated waveforms (achieving 96.77% accuracy), the technique falls apart when being used on real-world measured waveforms. This means that the current method of simulating waveforms does not fully represent real-world conditions. Further research is needed in this area. Subsequent PSO and DBSCAN analyses on the output of Envelope Analysis can return moderately good results, successfully identifying the correct number of PD sources (6 out of 6), with a high localization Euclidean error of ~4m.

The other clustering method, WD + PCA does not accurately represent the signal, as it is highly sensitive to the distance travelled and the noise level. However, further

analysis on the output of WD showed that the coefficient cA5 had promising ability to distinguish between different PD types.

Both Envelope Analysis and WD + PCA reduce the time and frequency information of the waveforms to scalar values, such as the difference coefficient and normalized component energies. This makes it difficult to distinguish between different waveforms, as their shapes may be different from one another, but the scalar values returned from the clustering methods are similar. In the future, an innovative method is needed to quantize the shape information of the signals, such as the number of peaks, location of each peak, etc.

Therefore, the work for Semester 2 can be summarized below:

- i. Research more accurate ways to model PD UHF discharge, possibly using COMSOL.
- ii. Implementing the remaining clustering/feature extraction methods: K-means, Gaussian mixture model, deep learning, selected bispectra and statistical analysis.
- iii. Innovating by combining and modifying existing feature extraction methodologies, aiming to accurately quantize signal shape information.
- iv. Aiming to achieve >90% clustering accuracy (Stage 1 Accuracy) for measured PD waveforms.

CHAPTER 5: CONCLUSION

5.1 Conclusion

This study successfully developed a comprehensive methodology for the simultaneous localization of multiple Partial Discharge (PD) sources, addressing the complex challenge of distinguishing overlapping signals in high-voltage environments. By integrating a Digital Twin simulation environment with advanced signal processing techniques, the study evaluated the efficacy of various feature extraction and clustering algorithms.

The simulation results demonstrated that Envelope Analysis combined with Particle Swarm Optimization (PSO) and DBSCAN was the most effective approach in the digital domain, achieving a clustering accuracy of 96.77% and successfully identifying 6 out of 6 simulated PD sources, despite the high Euclidean error of ~4m. Conversely, the Wavelet Decomposition (WD) with Principal Component Analysis (PCA) method proved highly sensitive to signal attenuation and noise, failing to consistently cluster sources based on distance.

However, the application of these methods to real-world measured waveforms revealed a significant gap between the digital twin and physical reality. While the simulation accurately modeled ideal conditions, the experimental data from void and incision defects showed that scalar features (such as the difference coefficient in Envelope Analysis) were insufficient for distinguishing between physically similar PD types in a noisy environment. Despite these challenges, the study confirmed that the cA5 coefficient from Wavelet Decomposition holds promise for distinguishing different PD types, providing a clear direction for refining feature extraction strategies.

5.2 Impact of Work to Environment and Society

The development of accurate Partial Discharge (PD) localization techniques has profound implications for both environmental sustainability and societal stability. From an environmental perspective, high-voltage equipment failures often lead to catastrophic events such as transformer explosions or leaks of insulating fluids. For instance, the leakage of Sulphur Hexafluoride (SF₆), a potent greenhouse gas commonly used in switchgear, significantly contributes to climate change. By enabling early detection and precise localization of insulation defects, this research facilitates preventative maintenance, thereby reducing the risk of equipment rupture and the subsequent release of hazardous materials into the environment. Societally, the reliability of the power grid is paramount. Unexpected power outages caused by insulation breakdown can disrupt critical infrastructure, including hospitals, transportation networks, and communication systems, leading to economic loss and public safety hazards. The capability to separate and localize multiple PD sources simultaneously allows utility providers to monitor the health of the grid more effectively, ensuring uninterrupted power supply and enhancing the overall resilience of the energy infrastructure for the community.

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